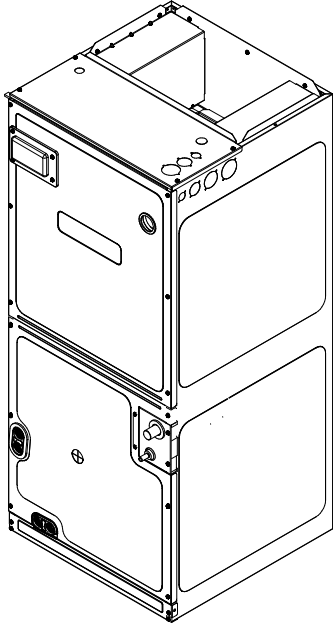


AMST***U1300**

AIR HANDLERS INSTALLATION & OPERATING INSTRUCTIONS

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P/N: IO-4011H Date: April 2026



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WARNING

ONLY PERSONNEL THAT HAVE BEEN TRAINED TO INSTALL, ADJUST, SERVICE, MAINTENANCE OR REPAIR (HEREINAFTER, "SERVICE") THE EQUIPMENT SPECIFIED IN THIS MANUAL SHOULD SERVICE THE EQUIPMENT.

THIS EQUIPMENT IS NOT INTENDED FOR USE BY PERSONS (INCLUDING CHILDREN) WITH REDUCED PHYSICAL, SENSORY OR MENTAL CAPACITIES, OR LACK OF EXPERIENCE AND KNOWLEDGE, UNLESS THEY HAVE BEEN GIVEN SUPERVISION OR INSTRUCTION CONCERNING USE OF THE APPLIANCE BY A PERSON RESPONSIBLE FOR THEIR SAFETY.

CHILDREN SHOULD BE SUPERVISED TO ENSURE THAT THEY DO NOT PLAY WITH THE EQUIPMENT.

THE MANUFACTURER WILL NOT BE RESPONSIBLE FOR ANY INJURY OR PROPERTY DAMAGE ARISING FROM IMPROPER SUPERVISION, SERVICE OR SERVICE PROCEDURES. IF YOU SERVICE THIS UNIT, YOU ASSUME RESPONSIBILITY FOR ANY INJURY OR PROPERTY DAMAGE WHICH MAY RESULT. IN ADDITION, IN JURISDICTIONS THAT REQUIRE ONE OR MORE LICENSES TO SERVICE THE EQUIPMENT SPECIFIED IN THIS MANUAL, ONLY LICENSED PERSONNEL SHOULD SERVICE THE EQUIPMENT. IMPROPER SUPERVISION, INSTALLATION, ADJUSTMENT, SERVICING, MAINTENANCE OR REPAIR OF THE EQUIPMENT SPECIFIED IN THIS MANUAL, OR ATTEMPTING TO INSTALL, ADJUST, SERVICE OR REPAIR THE EQUIPMENT SPECIFIED IN THIS MANUAL WITHOUT PROPER SUPERVISION OR TRAINING MAY RESULT IN PRODUCT DAMAGE, PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



WARNING

DO NOT BYPASS SAFETY DEVICES.



RECOGNIZE THIS SYMBOL AS A SAFETY PRECAUTION

NOTE: SPECIFICATIONS AND PERFORMANCE DATA LISTED HEREIN ARE SUBJECT TO CHANGE WITHOUT NOTICE.

NOTE: THE EQUIPMENT IS ONLY APPROVED FOR USE WITH R-32 REFRIGERANT.

NOTE: THE INSTALLER MUST REFER TO SECTION 13 FOR THE MAXIMUM POSSIBLE REFRIGERANT CHARGE AND MINIMUM (AIR CONDITIONED) ROOM AREA WHICH MUST BE RECORDED ON THE SERIAL PLATE.



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1 Important Safety Instructions

The following symbols and labels are used throughout this manual to indicate immediate or potential safety hazards. It is the owner's and installer's responsibility to read and comply with all safety information and instructions accompanying these symbols. Failure to heed safety information increases the risk of personal injury, property damage, and/or product damage.



WARNING

HIGH VOLTAGE

DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



WARNING

TO PREVENT THE RISK OF PROPERTY DAMAGE, PERSONAL INJURY, OR DEATH, DO NOT STORE COMBUSTIBLE MATERIALS OR USE GASOLINE OR OTHER FLAMMABLE LIQUIDS OR VAPORS IN THE VICINITY OF THIS UNIT.



WARNING

THIS PRODUCT IS FACTORY-SHIPED FOR USE WITH 208V/240V/1PH/60HZ ELECTRICAL POWER SUPPLY. **DO NOT** RECONFIGURE THIS AIR HANDLER TO OPERATE WITH ANY OTHER POWER SUPPLY.



WARNING

TO AVOID PROPERTY DAMAGE, PERSONAL INJURY OR DEATH DUE TO ELECTRICAL SHOCK, THIS UNIT **MUST** HAVE AN UNINTERRUPTED, UNBROKEN ELECTRICAL GROUND. THE ELECTRICAL GROUND CIRCUIT MAY CONSIST OF AN APPROPRIATELY SIZED ELECTRICAL WIRE CONNECTING THE GROUND LUG IN THE UNIT CONTROL BOX TO THE BUILDING ELECTRICAL SERVICE PANEL. OTHER METHODS OF GROUNDING ARE PERMITTED IF PERFORMED IN ACCORDANCE WITH THE NATIONAL ELECTRIC CODE (NEC) / AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) / NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) 70 AND LOCAL/STATE CODES. IN CANADA, ELECTRICAL GROUNDING IS TO BE IN ACCORDANCE WITH THE CANADIAN ELECTRIC CODE (CSA) C22.1.



CAUTION

WHEN INSTALLING OR SERVICING THIS EQUIPMENT, SAFETY CLOTHING, INCLUDING HAND AND EYE PROTECTION, IS STRONGLY RECOMMENDED. IF INSTALLING IN AN AREA THAT HAS SPECIAL SAFETY REQUIREMENTS (HARD HATS, ETC.), OBSERVE THESE REQUIREMENTS.



WARNING

A REFRIGERANT LEAK DETECTION SYSTEM IS INSTALLED. THE UNIT MUST BE POWERED ON AT ALL TIMES EXCEPT FOR SERVICE.



DANGER PELIGRO



CARBON MONOXIDE POISONING HAZARD

Special Warning for Installation of Furnace or Air Handling Units in Enclosed Areas such as Garages, Utility Rooms or Parking Areas

Carbon monoxide producing devices (such as an automobile, space heater, gas water heater, etc.) should not be operated in enclosed areas such as unventilated garages, utility rooms or parking areas because of the danger of carbon monoxide (CO) poisoning resulting from the exhaust emissions. If a furnace or air handler is installed in an enclosed area such as a garage, utility room or parking area and a carbon monoxide producing device is operated therein, there must be adequate, direct outside ventilation.

This ventilation is necessary to avoid the danger of CO poisoning which can occur if a carbon monoxide producing device continues to operate in the enclosed area. Carbon monoxide emissions can be (re)circulated throughout the structure if the furnace or air handler is operating in any mode.

CO can cause serious illness including permanent brain damage or death.

B10259-216

RIESGO DE INTOXICACIÓN POR MONÓXIDO DE CARBONO

Advertencia especial para la instalación de calentadores y manejadoras de aire en áreas cerradas como estacionamientos o cuartos de servicio.

Los equipos o aparatos que producen monóxido de carbono (tal como automóvil, calentador de gas, calentador de agua por medio de gas, etc) no deben ser operados en áreas cerradas debido al riesgo de envenenamiento por monóxido de carbono (CO) que resulta de las emisiones de gases de combustión. Si el equipo o aparato se opera en dichas áreas, debe existir una adecuada ventilación directa al exterior.

Esta ventilación es necesaria para evitar el peligro de envenenamiento por CO, que puede ocurrir si un dispositivo que produce monóxido de carbono sigue operando en el lugar cerrado.

Las emisiones de monóxido de carbono pueden circular a través del aparato cuando se opera en cualquier modo.

El monóxido de carbono puede causar enfermedades severas como daño cerebral permanente o muerte.

B10259-216

RISQUE D'EMPOISONNEMENT AU MONOXYDE DE CARBONE

Avertissement special au sujet de l'installation d'appareils de chauffage ou de traitement d'air dans des endroits clos, tels les garages, les locaux d'entretien et les stationnements.

Evitez de mettre en marche les appareils produisant du monoxyde de carbone (tels que les automobile, les appareils de chauffage autonome, etc.) dans des endroits non ventilés tels que les d'empoisonnement au monoxyde de carbone. Si vous devez faire fonctionner ces appareils dans un endroit clos, assurez-vous qu'il y ait une ventilation directe provenant de l'exterieur.

Cette ventilation est nécessaire pour éviter le danger d'intoxication au CO pouvant survenir si un appareil produisant du monoxyde de carbone continue de fonctionner au sein de la zone confinée.

Les émissions de monoxyde de carbone peuvent être redirigées dans les endroits clos, si l'appareil de chauffage ou de traitement d'air sont en marche.

Le monoxyde de carbone peut causer des maladies graves telles que des dommages permanents au cerveau et même la mort.

B10259-216

2 Shipping Inspection

Always transport the unit upright; laying the unit on its side or top during transit may cause equipment damage. The installer should inspect the product upon receipt for shipping damage and subsequent investigation is the responsibility of the carrier. The installer must verify the model number, specifications, electrical characteristics, and accessories are correct prior to installation. The distributor or manufacturer will not accept claims from dealers for transportation damage or installation of incorrectly shipped units.

2.1 PARTS

Inspect the unit to verify all required components are present and intact. Report any missing components immediately to the manufacturer or to the distributor. Use only factory authorized replacement parts (see Section 4). Make sure to include the full product model number and serial number when reporting and/or obtaining service parts.

2.2 HANDLING

Use caution when transporting/carrying the unit. Do not carry the unit with hooks or sharp objects. The preferred method of carrying the unit after arrival at the job site is to carry by a two-wheel hand truck from the back or sides. If carrying by hand, carry at the cabinet corners with two people. Avoid holding the unit by the piping.

2.3 SHIPPING MATERIAL REMOVAL

IMPORTANT: All Shipping Material used to protect the equipment, and the equipment's components, during transit should be removed before final installation.

2.3.1 BLOWER SUPPORT BRACKET REMOVAL

A shipping bracket (Figure 1) has been factory installed in this Air Handler to prevent any unintentional damage to the blower deck assembly during transit. The shipping bracket should be removed prior to installation. To remove the shipping bracket, remove the two (2) screws that secure the bracket to the equipment. Screw (A) is removed from the control deck and Screw (B) is removed from the inside corner post. The bracket can be discarded after removal.



WARNING

IF ANY HOT WORK IS TO BE CONDUCTED ON THE REFRIGERATING EQUIPMENT OR ANY ASSOCIATED PARTS, APPROPRIATE FIRE EXTINGUISHING EQUIPMENT SHALL BE AVAILABLE ON HAND. HAVE A DRY POWDER OR CO2 FIRE EXTINGUISHER ADJACENT TO THE HOT WORK AREA.

ENSURE THAT THE AREA IS IN THE OPEN OR THAT IT IS ADEQUATELY VENTILATED BEFORE BREAKING INTO THE SYSTEM OR CONDUCTING ANY HOT WORK. A DEGREE OF VENTILATION SHALL CONTINUE DURING THE PERIOD THAT THE WORK IS CARRIED OUT. THE VENTILATION SHOULD SAFELY DISPERSE ANY RELEASED REFRIGERANT AND PREFERABLY EXPEL IT EXTERNALLY INTO THE ATMOSPHERE.

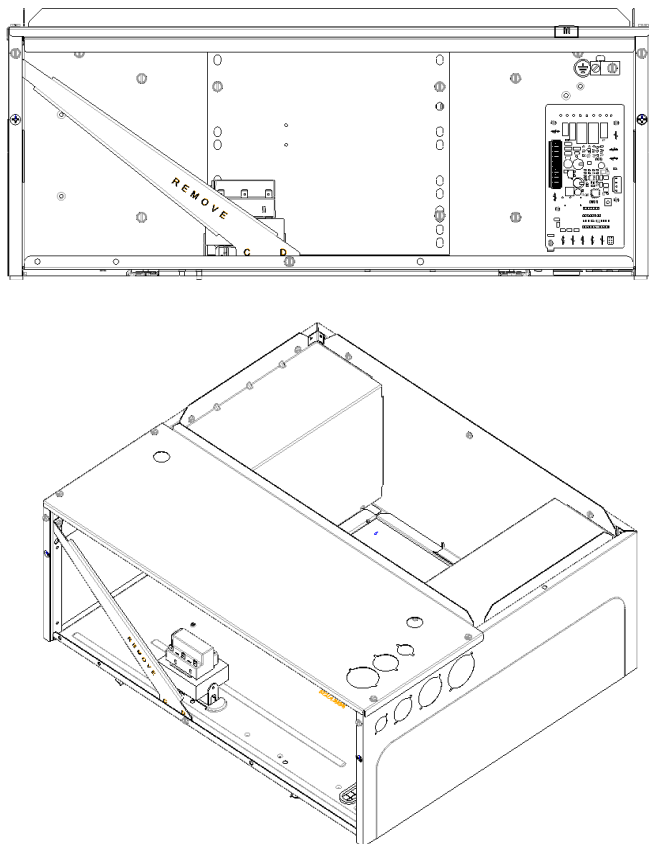


FIGURE 1
BLOWER SUPPORT BRACKET REMOVAL

3 Codes & Regulations

This product is designed and manufactured to comply with national codes such as UL60335 2-40, ASHRAE 15, ASHRAE 15.2, IAPMO Uniform Mechanical Code, ICC International Mechanical Code, or CSA B52. Installation in accordance with such codes and/or prevailing local codes/regulations is the responsibility of the installer. The manufacturer assumes no responsibility for equipment installed in violation of any codes or regulations.

The United States Environmental Protection Agency (EPA) has issued various regulations regarding the introduction and disposal of refrigerants. Failure to follow these regulations may harm the environment and can lead to the imposition of substantial fines. Should you have any questions please contact the local office of the EPA and / or refer to EPA's website www.epa.gov.

4 Replacement Parts

When reporting shortages or damages, or ordering repair parts, give the complete product model and serial numbers as on the unit's Serial and Rating Plate. Replacement parts for this product are available through your contractor or local distributor.

5 Pre-Installation Considerations

5.1 PREPARATION

Keep this document with the unit. Carefully read all instructions for the installation prior to installing product. Make sure each step or procedure is understood and any special considerations are taken into account before starting installation. Assemble all tools, hardware and supplies needed to complete the installation. Some items may need to be purchased locally. Make sure everything needed to install the product is on hand before starting.

5.2 SYSTEM MATCHES

The entire system (combination of indoor and outdoor sections) must be manufacturer approved and Air-Conditioning, Heating, and Refrigeration Institute (AHRI) listed. **NOTE:** Installation of unmatched systems is not permitted. Damage or repairs due to installation of unmatched systems is not covered under the warranty.

5.3 INTERCONNECTING TUBING

Give special consideration to minimize the length of refrigerant tubing when installing air handlers. Refer to Remote Cooling / Heat Pump Service Manual, and TP-110* Long Line Set Application R-32 for tubing guidelines. If possible, allow adequate length of tubing such that the coil may be removed (for inspection or cleaning services) from the cabinet without disconnecting the tubing.

5.4 CLEARANCES



WARNING

WHEN INSTALLED IN A ROOM WITH AN AREA LESS THAN THAT OUTLINED IN TABLE 6, THAT ROOM SHALL BE WITHOUT CONTINUOUSLY OPERATING OPEN FLAMES (FOR EXAMPLE AN OPERATING GAS APPLIANCE) OR OTHER POTENTIAL IGNITION SOURCES (FOR EXAMPLE AN OPERATING ELECTRIC HEATER, HOT SURFACES).



WARNING

AUXILIARY DEVICES WHICH MAY BE A POTENTIAL IGNITION SOURCE ARE NOT TO BE INSTALLED IN THE DUCT WORK. SUCH POTENTIAL IGNITION SOURCES INCLUDE HOT SURFACES WITH A TEMPERATURE EXCEEDING 650°C AND ELECTRIC SWITCHING DEVICES. FOR EXAMPLE ELECTRONIC AIR FILTERS ARE A POTENTIAL IGNITION SOURCE.



WARNING

THE ONLY AUXILIARY DEVICES APPROVED BY THE APPLIANCE MANUFACTURER OR DECLARED SUITABLE WITH THE REFRIGERANT SHALL BE INSTALLED IN CONNECTING DUCTWORK. IF AN EXISTING DEVICE IS NOT APPROVED BY THE APPLIANCE MANUFACTURER FOR USE WITH THE REFRIGERANT THE DEVICE MUST BE REMOVED FROM THE DUCTWORK.

The unit clearance from a combustible surface may be 0". However, service clearance must take precedence. A minimum of 24" in front of the unit for service clearance is required. Additional clearance on one side or top will be required for electrical wiring connections. Consult all appropriate regulatory codes prior to determining final clearances. When installing this unit in an area that may become wet (such as crawl spaces), elevate the unit with a sturdy, non-porous material. In installations that may lead to physical damage (i.e. a garage) it is advised to install a protective barrier to prevent such damage. Always install units such that a positive slope in condensate line (1/4" per foot) is allowed.

Refrigerant lines must be routed depending on configuration of unit to maintain the required 24" minimum clearance for service. Consult all appropriate regulatory codes prior to determining final clearances. In installations that may lead to physical damage (i.e. a garage) it is advised to install a protective barrier to prevent such damage. Always install units such that a positive slope in condensate line (1/4" per foot) is allowed.

Further, any joints made in the installation between parts of the refrigerating system must be accessible for maintenance purposes.

5.5 HORIZONTAL APPLICATIONS

If installed above a finished living space, a secondary drain pan with float switch (as required by many building codes), must be installed under the entire unit and its condensate drain line must be routed to a location such that the user will see the condensate discharge.

5.6 ACCESS

This unit should be installed in a manner so that it is not accessible to the general public.

6 Installation Location

NOTE: These air handlers are designed for indoor installation only at a max altitude of 10,500 feet above sea level or a min altitude of -184 feet below sea level.

If the unit is located in an unconditioned area with high ambient temperature and/or high humidity, the air handler may be subject to nuisance sweating of the casing. On these installations, a wrap of 2" fiberglass insulation with a vapor barrier is recommended. A secondary drain pan below the unit is also recommended to protect the installation site.

This product line may be installed in one of the upflow, downflow, horizontal left or horizontal right orientations as shown in Figures 5, 6, 7 and 8. The unit may be installed in horizontal left orientation as shipped (refer to specific sections for more information).

6.1 UPFLOW AND DOWNFLOW INSTALLATION (FIGURE 10 & 11)

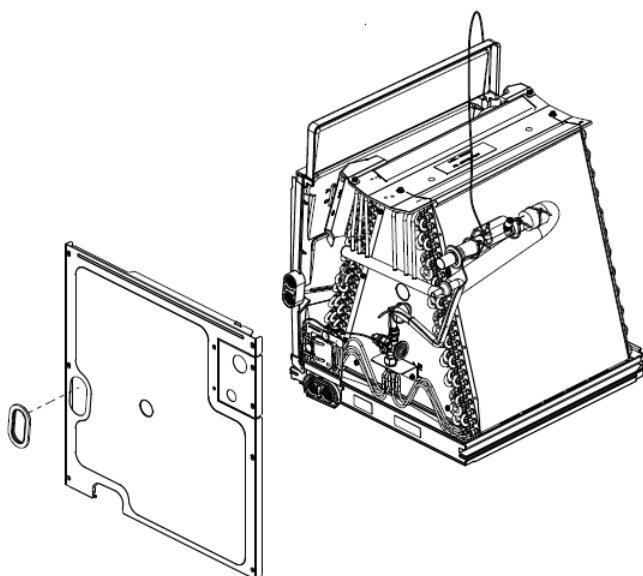
1. Remove blower access panel, coil access panel, and tubing access panel as shown in Figure 12 & 15. Remove and retain the seven (7) screws securing the coil access panel to the cabinet and the six (6) screws securing the blower access panel to the cabinet.
2. Disconnect the sensor wire from the A2L PCB. Slide the coil assembly out from the cabinet. Use the drain pan to pull the assembly from the cabinet.
IMPORTANT: Do not use manifolds, copper lines, or the flowrator to pull the coil assembly out. Failure to do so may result in braze joint damage and leaks.
3. Removal of the center support is required on units with 21" wide cabinet. Remove and retain the two (2) screws that secure the center support to the cabinet. Remove the center support. Reference Figure 10
4. **The horizontal drip shield, side drain pan, and drain pan extension should be removed. Follow the "Side Drain Pan and Extension Removal Instructions".**
5. **For Upflow Installation:**
 - a. Using the drain pan to hold the coil assembly, slide the coil assembly back into the cabinet.
 - b. Reinstall the center support (if removed) using the two (2) screws removed in Step 3.
 - c. Reconnect the sensor wire to the A2L PCB.
 - d. Reinstall Coil Access panel, Tubing Access and Blower Access panel removed in Step 1. Reference Figure 11
6. **For Downflow Installation:**
 - a. Position the unit in the downflow position.
 - b. Using the drain pan to hold the coil assembly, slide the coil assembly back into the cabinet on the downflow brackets as shown in Figure 13.
 - c. Reinstall the center support (if removed) using the two (2) screws removed in Step 3.
 - d. Reinstall Coil Access panel, Tubing Access and Blower Access panel removed in Step 1. Reference Figure 14.

Side Drain Pan and Extension Removal Instructions

Refer to Figure 4, remove the two (2) screws that secure the drip shield support brackets to the condensate collectors (front and back). Unsnap the side drain pan from the bottom drain pan using a screwdriver or any small lever. The side drain pan, drip shield brackets and the drain pan extension may now be removed. From Figure 4, drain port labeled (A) is the primary drain for this application and condensate drain line must be attached to this drain port. Drain port (a) is for the secondary drain line (if used). When the side drain pan is removed, the drain port opening in the access panel must be covered by the accessory drain port plug (DPK1) as shown in Figure 2.

Drain Port Plug		
Kit Number	Description	Application
DPK1	Side Drain Port Plug	All Models

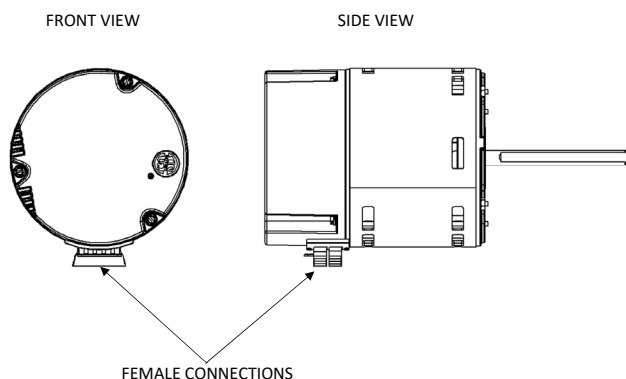
DRAIN PORT PLUG KIT
TABLE 8



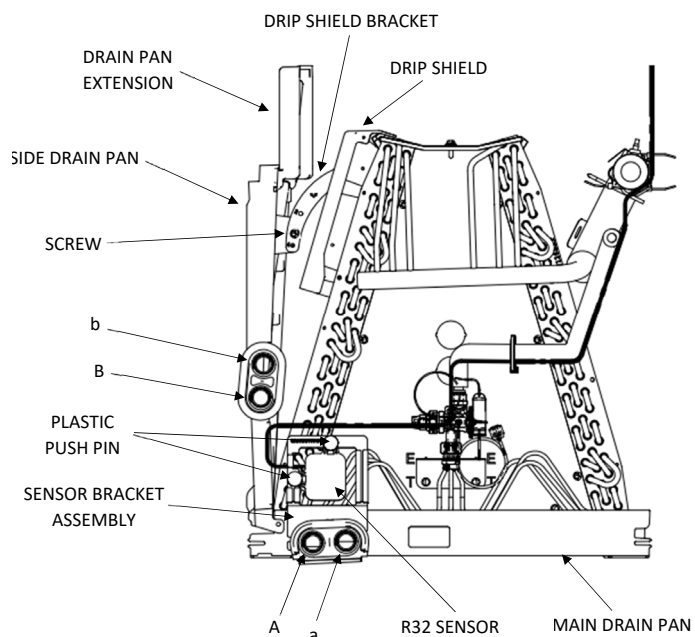
DRAIN PAN
FIGURE 2

6.1.1 MOTOR ORIENTATION

If the motor is not oriented with the connections on the casing pointing down, water could collect in the motor and cause premature failure. Loosen the motor mount and rotate the motor as shown in Figure 3.



MOTOR ORIENTATION
FIGURE 3



SIDE DRAIN PAN REMOVAL
FIGURE 4

6.2 HORIZONTAL LEFT INSTALLATION (FIGURE 7)

No field modifications are permissible for this application. Install unit as shown in Figure 7 apart from one mentioned in 10.2.1

Drain port labeled (B) in Figure 4 is the primary drain for this application and condensate drain line must be attached to this drain port. Drain port (b) is for the secondary drain line (if used).

Remove red plugs from side drain pan before connecting condensate drain pipes. Use removed plug to close drain ports on vertical drain pan.

6.2.1 INSTRUCTION TO RELOCATE SENSOR BRACKET ASSEMBLY FROM UPFLOW TO HORIZONTAL LEFT

1. Take off the blower access panel and coil access panel.
2. Take off the drain port gaskets on main drain pan and side drain pan
3. Take off the R-32 sensor bracket assembly
4. Put R-32 sensor bracket assembly to the side drain port correctly as shown in Figure 7. The "FRONT" print on sensor bracket should be in the front
5. Place gaskets back to the drain ports correctly. The "FRONT" print on the gaskets should be in the front.
6. Sensor wiring should be rerouted as shown in Figure 7.
7. Reassemble the blower access panel and coil access panel to the unit.

6.3 HORIZONTAL RIGHT INSTALLATION (FIGURE 8)

NOTE: For AMST24BU1300 only: If installing a filter, an external filter must be used when installing the unit in Horizontal Right. A filter will not fit on the internal filter rails in these applications.**

1. Before inverting the air handler, remove blower access panel and coil access panel. The coil access panel and tubing panel may remain screwed together during this procedure. Refer to Figure 2. Remove and retain the seven (7) screws securing the coil access panel to the cabinet and the six (6) screws securing the blower access panel to the cabinet as shown in Figure 12 & 15.
2. Removal of the center support is required on units with 21" wide cabinet. Remove and retain the two (2) screws that secure the center support to the cabinet. Remove the center support. Referring to Figure 10.
3. Slide the coil assembly out from the cabinet. Use the drain pan to pull the assembly from the cabinet.
NOTE: Do not use manifolds, copper lines, or the flowrator to pull the coil assembly out. Failure to do so may result in braze joint damage and leaks.
4. Position the unit in the downflow position. Refer to Figure 6.
5. Remove side drain pan extension if present.
6. Using the drain pan to hold the coil assembly, slide the coil assembly back into the cabinet on the downflow brackets as shown in Figure 6.
7. Reinstall the center support (if removed) using the two (2) screws removed in Step 2.
8. Reinstall the coil access panels and reinstall blower access panel removed in Step 1 as shown in Figure 14.
9. Drain Connections for Horizontal Right Installation
 - a. Drain port labeled (B) in Figure 4 is the primary drain for this application and condensate drain line must be attached to this drain port. Drain port (b) is for the secondary drain line (if used).
 - b. Remove red plugs from side drain pan before connecting condensate drain pipes. Use removed plug to close drain ports on vertical drain pan.

NOTE: If removing only the coil access panel from the unit, the filter access panel must be removed first. Failure to do so will result in panel damage.

Instructions to relocate Sensor bracket assembly from Upflow to Horizontal Right

1. Take off the blower access panel and coil access panel.
2. Take off the drain port gaskets on main drain pan and side drain pan
3. Disconnect the sensor wire from the A2L PCB.
4. Take off the R-32 sensor bracket assembly
5. Put R-32 sensor bracket assembly to the side drain port correctly as shown in Figure 8. The "FRONT" print on sensor bracket should be in the front
6. Place gaskets back to the drain ports correctly. The "FRONT" print on the gaskets should be in the front.

7. Reconnect the sensor wire to the A2L PCB.
8. Sensor wiring should be rerouted as shown in Figure 8.
9. Reassemble the blower access panel and coil access panel to the unit.

6.4 HUMID ENVIRONMENT INSTALLATIONS

NOTE: Each service kit mentioned in this instruction is available through your local distributor. The kits are not supplied with the air handler and must be used to assist with condensation management in humid environment applications. The individual installation instructions provided with each kit must be followed.

6.4.1 ALL INSTALLATIONS – HUMID ENVIRONMENTS

Installations in areas where the return air environment sees humidity levels above 65% relative humidity, a High Humidity Kit (HHK) must be used. See Table 1 for Model and Kit assignment.

HHK0004	HHK0005	HHK0006	HHK0008
AMST24B AMST30B AMST36B	AMST36C AMST42C	AMST48C	AMST60D

TABLE 1

6.4.1.2 DOWNFLOW INSTALLATIONS – HUMID ENVIRONMENTS

To prevent the coil pan "sweating", the mandatory Downflow Kit (DFK) must also be used in addition to the High Humidity Kit (HHK). See Table 2 for the correct DFK for each model.

DFKE-02	DFKE-03
AMST24B AMST30B AMST36B AMST36C AMST42C	AMST48C AMST60D

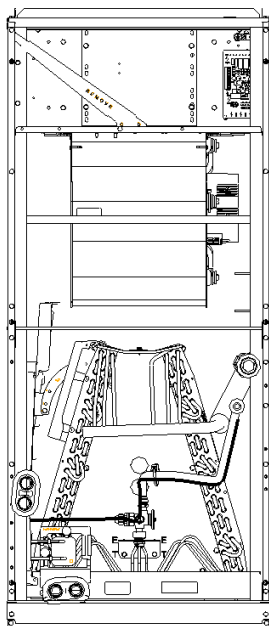
TABLE 2

6.4.1.3 HORIZONTAL INSTALLATIONS – HUMID ENVIRONMENTS

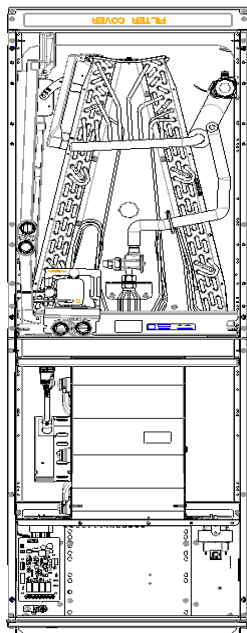
In addition to the High Humidity Kit (HHK), in applications where the air handler is installed in the horizontal position (Left or Right), and the return air environment sees humidity levels above 65% relative humidity, a Condensate Management Kit (CMK) must be used for field application. Applicable Kit and Model listing can be found in Table 3.

CMK0018	CMK0019	CMK0020
AMST24B AMST36C	AMST30B AMST36B AMST42C	AMST48C AMST60D

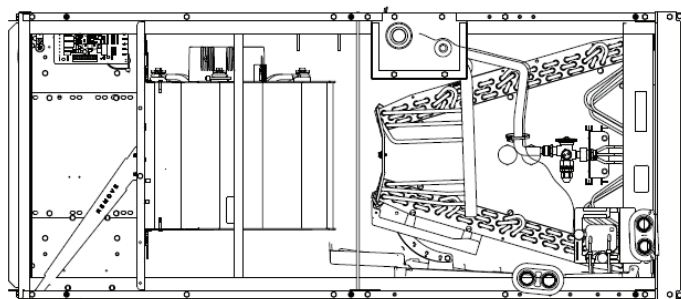
TABLE 3



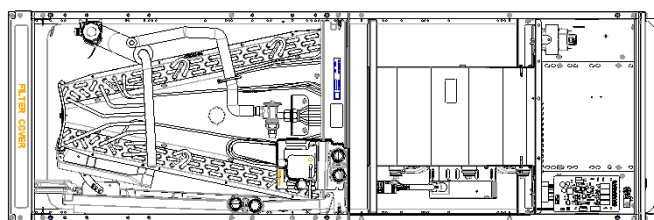
**UPFLOW
FIGURE 5**



**DOWNFLOW
FIGURE 6**



**HORIZONTAL LEFT
FIGURE 7**



**HORIZONTAL RIGHT
FIGURE 8**

7 Refrigerant Lines



WARNING

DO NOT USE MEANS TO ACCELERATE THE DEFROSTING PROCESS OR TO CLEAN, OTHER THAN THOSE RECOMMENDED BY THE MANUFACTURER.

THE APPLIANCE SHALL BE STORED IN A ROOM WITHOUT CONTINUOUSLY OPERATING IGNITION SOURCES (FOR EXAMPLE: OPEN FLAMES, AN OPERATING GAS APPLIANCE OR AN OPERATING ELECTRIC HEATER.) DO NOT PIERCE OR BURN. BE AWARE THAT REFRIGERANTS MAY NOT CONTAIN AN ODOR.



WARNING

THIS PRODUCT IS FACTORY-SHIPED WITH DRY NITROGEN MIXTURE GAS UNDER PRESSURE. USE APPROPRIATE SERVICE TOOLS AND FOLLOW THESE INSTRUCTIONS TO PREVENT INJURY.



WARNING

WORK SHALL BE UNDERTAKEN UNDER A CONTROLLED PROCEDURE TO MINIMIZE THE RISK OF FLAMMABLE GAS OR VAPOR BEING PRESENT WHILE THE WORK IS BEING PERFORMED.



WARNING

ONLY BRAZING TECHNIQUES AND APPROVED MECHANICAL JOINTS SHOULD BE USED TO CONNECT REFRIGERANT TUBING CONNECTIONS, NON-APPROVED MECHANICAL CONNECTORS AND OTHER METHODS ARE NOT PERMITTED IN THIS SYSTEM CONTAINING A2L REFRIGERANT. APPROVED MECHANICAL JOINTS WILL BE DETAILED IN THE PRODUCT'S SPECIFICATION SHEETS.

NOTE: Refrigerant tubing must be routed to allow adequate access for servicing and maintenance of the unit.

Do not install the air handler in a location that violates the instructions provided with the condenser. If the unit is located in an unconditioned area with high ambient temperature and/or high humidity, the air handler may be subject to nuisance sweating of the air handler cabinet. On these installations, a wrap of 2" fiberglass insulation with a vapor barrier is recommended.

Follow standards related to ASHRAE 15, ASHRAE 15.2, IAPMO Uniform Code or CSA B52 during any installation. Piping must be enclosed in compliance with national and local codes.

Field-made refrigerant joints indoors shall be tightness tested. The test method shall have a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0.25 times the maximum allowable pressure. No leak shall be detected (Should be leak tight).

The indoor equipment and pipes must be securely mounted and guarded such that accidental rupture of equipment cannot occur from such events as moving furniture or reconstruction activities. Field-made refrigerant joints indoors shall be tightness tested.

Field pipework must be pressure tested with an inert gas and then vacuum tested prior to refrigerant charging according to the following requirements:

7.1 STANDING PRESSURE TEST (RECOMMENDED BEFORE VACUUM)

Pressurize the system to the maximum allowable pressure listed in the serial plate using dry nitrogen or dry helium. Allow the pressure to stabilize and hold for at least 15 minutes. The system is considered leak-free if the pressure does not drop below the selected maximum allowable pressure. If, after 15 minutes, the pressure drops, it implies a leak in the system. Proceed with identifying and sealing the leak and repeating the Standing Pressure Test. Proceed to system evacuation using the Deep Vacuum Method. Leak test the system using dry nitrogen or dry helium and soapy water to identify leaks. **NO REFRIGERANT SHALL BE USED FOR PRESSURE TESTING TO DETECT LEAKS.** Proceed to system evacuation using the Deep Vacuum Method.

No refrigerant shall be used for pressure testing to detect leaks.

Proceed to system evacuation using the Deep Vacuum Method.

7.2 DEEP VACUUM METHOD

The Deep Vacuum Method requires a vacuum pump rated for 500 microns or less. This method effectively and efficiently ensures the system is free of non-condensable air and moisture. The Triple Evacuation Method is detailed in the Service Manual for this product model as an alternative. To expedite the evacuation procedure, it is recommended that the Schrader Cores be removed from the service valves using a core-removal tool.

1. Connect the vacuum pump, micron gauge, and vacuum-rated hoses to both service valves. Evacuation must use both service valves to eliminate system mechanical seals.
2. Evacuate the system to less than 500 microns.
3. Isolate the pump from the system and hold the vacuum for 10 minutes (minimum). Typically, pressure will rise slowly during this period. If the pressure rises to less than 1000 microns and remains steady, the system is considered leak-free; proceed to system charging and startup.
4. If pressure rises above 1000 microns but holds steady below 2000 microns, non-condensable air or moisture may remain, or a small leak may be present. Return to step 2: If the same result is achieved, check for leaks and repair. Repeat the evacuation procedure.

5. If pressure rises above 2000 microns, a leak is present. Check for leaks and repair them. Then, repeat the evacuation procedure.

7.3 TUBING SIZE/LENGTH

Give special consideration to minimize the length of refrigerant tubing when installing air handlers. Refer to Remote Cooling / Heat Pump Service Manual, and TP-110* Long Line Set Application R-32 for tubing guidelines. If possible, allow adequate length of tubing such that the coil may be removed (for inspection or cleaning services) from the cabinet without disconnecting the tubing.

7.4 TUBING PREPARATION

All cut ends are to be round, burr free, and clean. Failure to follow this practice increases the chances for refrigerant leaks. The suction line is spun closed and requires tubing cutters to remove the closed end.

Installation of pipe-work must be kept to a minimum. Pipe-work including piping material, pipe routing and installation must be protected from physical damage and shall not be installed in an unventilated space. Equipment piping in any occupied space shall be installed in such a way to protect against accidental damage in operation and service. Precautions must be taken to avoid excessive vibration or pulsation to refrigeration piping. Protection devices, piping and fittings shall be protected as far as possible against adverse environmental effects, for example, the danger of water collecting and freezing in relief pipes or the accumulation of dirt and debris.



WARNING

THIS UNIT MUST BE CHARGED WITH R-32 REFRIGERANT. ENSURE THAT SENSOR IS INSTALLED CORRECTLY AND IS OPERATIONAL.

7.5 BRAZING

Note: Use heat trap material or wet cloth behind braze joints to protect TXV and access fittings from overheating during brazing line set connections.

Braze joints should be made only with the connections provided external to the cabinet. Do not alter the cabinet nor braze inside the cabinet. To avoid overheating after brazing, quench all brazed joints with water or a wet rag.

This system must be charged with R32 refrigerant in order to operate safely.

When breaking into the refrigerant circuit to make repairs - or for any other purpose - conventional procedures shall be used. However, for flammable refrigerants it is important that best practice be followed, since flammability is a consideration. The following procedure shall be adhered to:

1. Safely remove refrigerant following local and national regulations;
2. Evacuate;
3. Purge the circuit with inert gas;

4. Evacuate;
5. Continuously flush or purge with inert gas when using flame to open circuit; and
6. Open the circuit.

For appliances containing flammable refrigerants, refrigerants purging shall be achieved by breaking the vacuum in the system with oxygen-free nitrogen and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum (optional for R32). This process shall be repeated until no refrigerant is within the system (optional for R32). When the final oxygen-free nitrogen charge is used, the system shall be vented down to atmospheric pressure to enable work to take place.

The outlet for the vacuum pump shall not be close to any potential ignition sources, and ventilation shall be available.



CAUTION

APPLYING TOO MUCH HEAT TO ANY TUBE CAN MELT THE TUBE. TORCH HEAT REQUIRED TO BRAZE TUBES OF VARIOUS SIZES MUST BE PROPORTIONAL TO THE SIZE OF THE TUBE. SERVICE PERSONNEL MUST USE THE APPROPRIATE HEAT LEVEL FOR THE SIZE OF THE TUBE BEING BRAZED.

7.6 THERMAL EXPANSION VALVE SYSTEM ADJUSTMENT

The following information for the indoor unit should be verified before attempting to charge system or adjust TXV if necessary.

1. Total static pressure is .5" WC or less.
2. Airflow is correct for installed unit.
3. Airflow tables are in the installation manual and Spec Sheet for Indoor Unit.
4. Complete airflow tables and charging information are in Service Manual.
5. The outdoor temperature must be 60°F or higher.
6. Set the room thermostat to COOL, fan switch to AUTO
7. Set the temperature control well below room temperature.

NOTE: Units matched with indoor coils equipped with a TXV should be charged by Subcooling only. Superheat can also be utilized to best verify charge levels with an adjustable TXV and make adjustments when needed in unique applications due to refrigerant line length, differences in height between the indoor and outdoor unit and refrigerant tubing sizes. These adjustments should only be performed by qualified service personnel.

Superheat adjustments should not be made until indoor ambient conditions have stabilized. This could take up to 24 hours depending on indoor temperature and humidity. Before checking superheat run the unit in cooling for 10 minutes or until refrigerant pressures stabilize. Use the following guidelines and methods to check unit operation and ensure that the refrigerant charge is within limits.

NOTE: Charge two stage units on low stage.

In addition to conventional charging procedures, the following requirements shall be followed:

1. Ensure that contamination of different refrigerants does not occur when using charging equipment. Hoses or lines shall be as short as possible to minimize the amount of refrigerant contained in them.
2. R-32 Refrigerant Cylinders shall be kept in an appropriate position.
3. Ensure that the refrigeration system is grounded (prior) to charging the system with refrigerant.
4. Label the system when charging is complete (if not already).
5. Extreme care shall be taken not to overfill the refrigerating system.



FIGURE 9

For adjustment, please refer to the illustration above. Carefully remove the nut without damaging the TXV. Rotate the TXV clockwise (right) to close it, ensuring the TXV is oriented as shown in the illustration. If it is not, adjust the orientation accordingly. Each adjustment should involve a rotation of 1/8 turn: clockwise (right) to close the TXV (as per illustration) and counterclockwise (left) to open it. The total combined adjustment must not exceed 1.5 turns in either direction from the default set point.

Prior to recharging the system, it shall be pressure-tested with the appropriate purging gas. The system shall be leaktested on completion of charging but prior to commissioning. A follow up leak test shall be carried out prior to leaving the site.

1. Purge gauge lines. Connect service gauge manifold to base-valve service ports.
2. Temporarily install a thermometer on the liquid line at the liquid line service valve and 4-6" from the compressor on the suction line. Ensure the thermometer makes adequate contact and is insulated for best possible readings. Use liquid line temperature to determine subcooling and vapor temperature to determine superheat.
3. If subcooling and superheat are low, adjust TXV superheat, then check subcooling as specified in Outdoor Installation and Instruction manual.

- a. If subcooling is low and superheat is high, add charge to raise subcooling to 8°F +/- 1°F, then check superheat.
- b. If subcooling and superheat are high, adjust TXV superheat, then check subcooling as specified in Outdoor Installation and Instruction manual.
4. After adjustments are complete replace red valve cap on adjustment stem and finger-tight the red valve cap.
5. Remove gauges and check the Schrader ports for leaks and tighten valve cores if necessary. Install caps finger tight.

For optimum system performance subcool and superheat should be matched as specified in Outdoor Installation and Instruction manual. Outdoor should be matched with approved AHRI combination, other approved system shall have subcooling set to 8°F ± 1°F and superheat set to 12°F ± 1°F.

NOTE: The minimum time between TXV adjustments shall be 10 minutes.

NOTE: In situations where the TXV must be removed and replaced or re-installed into the system, the TXV should be hand tightened first and then apply a half turn to fully set the TXV. No more than 20 ft-lb torque should be applied to the joints of the TXV.

NOTE: Do NOT adjust the charge based on suction pressure unless there is a gross undercharge.

SUBCOOL FORMULA = SAT. LIQUID LINE TEMP. - LIQUID LINE TEMP.

SUPERHEAT FORMULA = SUCT. LINE TEMP. - SAT. SUCT. TEMP.



CAUTION

EXCESSIVE TORQUE CAN CAUSE THE TXV RETAINING RING TO WEAKEN & CREATE A LEAK. THIS WILL MAKE IT DIFFICULT TO RE-INSTALL. THE PROPER TORQUE SETTINGS SHOULD BE CONFIGURED AS BELOW WHEN ADJUSTING:

OILED EQUALIZER SHOULD BE SET TO 10-12 FT/LBS.

OUTLET AND INLET CHAT FITTING SHOULD BE SET TO 15-20 FT/LBS.

NOTE: In situations where the TXV must be removed and replaced or re-installed into the system, the TXV should be hand tightened first and then apply a half turn to fully set the TXV. No more than 20 ft-lb torque should be applied to the joints of the TXV.

It is not allowed to install TXV outside the coil.

NOTE: Do NOT adjust the charge based on suction pressure unless there is a gross undercharge.

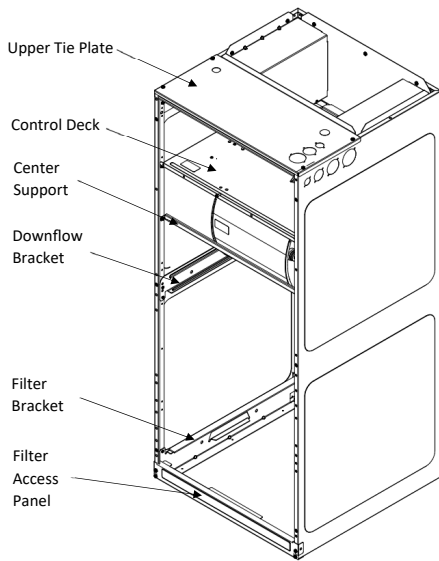
NOTE: The minimum time between TXV adjustments shall be 20 minutes.

7.7 LEAK DETECTION METHODS

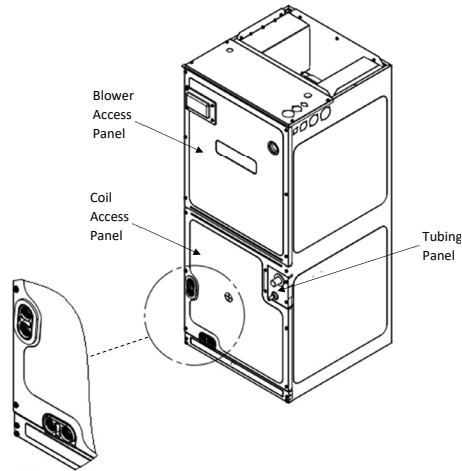
Under no circumstances shall potential sources of ignition be used in the searching for or detection of refrigerant leaks. A halide torch (or any other detector using a naked flame) shall not be used. If a leak is suspected, all naked flames shall be removed/extinguished. If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system, or isolated (by means of shut off valves) in a part of the system remote from the leak.

The following leak detection methods have been deemed acceptable, provided proper care is taken in their implementation:

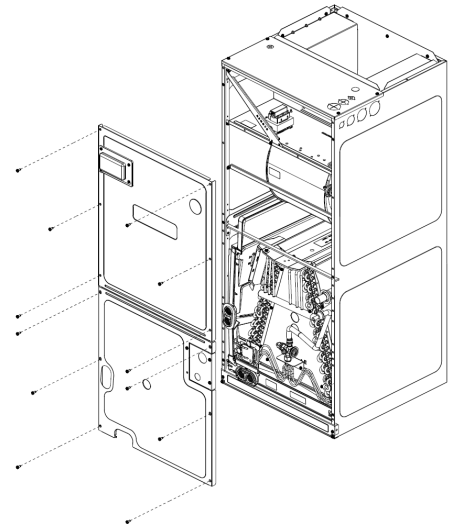
1. A2L compatible electronic leak detectors rated for use with R-32 refrigerant. Ensure the detection equipment is calibrated in a refrigerant-free area and that the leak detector itself is not a potential source of ignition. Leak detection equipment shall be set at a value of no more than 0.075 kg/m³ (1.2 lb/ft³).
2. Leak detection fluids - such as the bubble method or fluorescent method. Ensure that the use of detergents containing chlorine are avoided as the chlorine may react with the refrigerant and corroded the pipe-work.



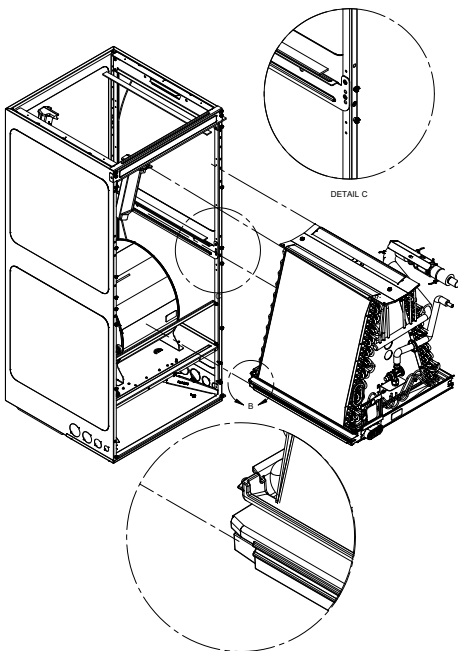
**INTERNAL PART TERMINOLOGY
FIGURE 10**



**EXTERNAL PART TERMINOLOGY
FIGURE 11**

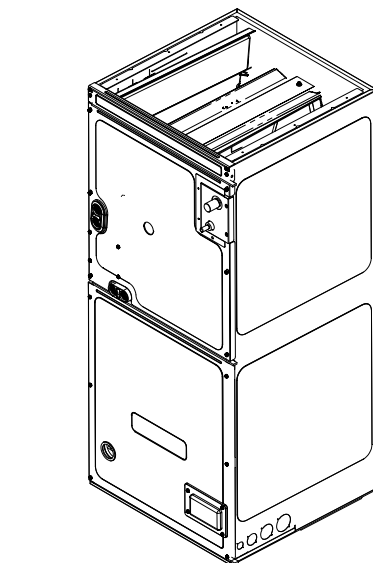


**EXPLODED VIEW OF EXTERNAL PART
TERMINOLOGY FOR UPFLOW
FIGURE 12**

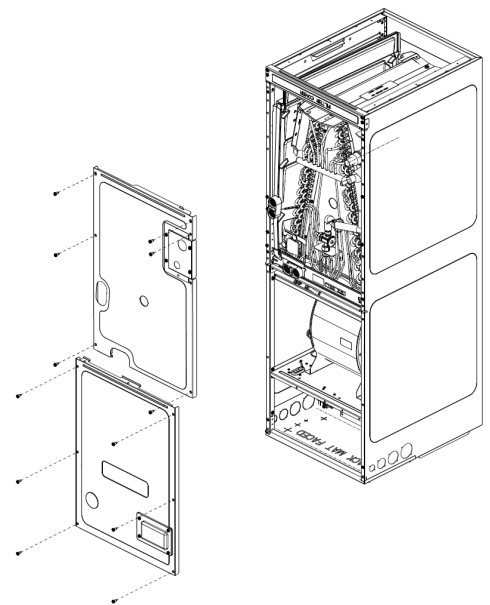


**COIL INSTALLATION FOR DOWNFLOW
FIGURE 13**

Important Note: Ensure coil slides on the rails along the groove provided on the drain pan side walls. Failure to do so will result in improper condensate drainage.



**ACCESS PANEL CONFIGURATION FOR
DOWNFLOW OR HORIZONTAL RIGHT
FIGURE 14**



**EXPLODED VIEW OF EXTERNAL PART
TERMINOLOGY FOR DOWNFLOW
FIGURE 15**

SATURATED SUCTION PRESSURE	
SUCTION PRESSURE	SATURATED SUCTION
PSIG	R-32
50	1
52	2
54	4
56	5
58	6
60	8
62	9
64	10
66	12
68	13
70	14
72	15
74	17
76	18
78	19
80	20
85	23
90	25
95	28
100	30
110	35
120	40
130	44
140	48
150	52
160	55
170	59

TABLE 4

SATURATED LIQUID PRESSURE	
LIQUID PRESSURE	SATURATED LIQUID TEMPERATURE °F
PSIG	R-32
200	68
210	71
220	74
225	76
235	78
245	81
255	84
265	86
275	88
285	91
295	93
305	95
325	100
355	106
375	110
405	116
415	117
425	119
435	121
445	123
475	128
500	132
525	135
550	139
575	143
600	146
625	150

TABLE 5

Minimum Mitigation Airflow Chart

Total System Refrigerant Charge	Maximum Possible Refrigerant Charge	Maximum Possible Refrigerant Charge	Minimum Air Conditioned Room Area	Minimum Air Conditioned Room Area	Minimum Required Mitigation Airflow	Minimum Required Mitigation Airflow
(oz)	(oz)	(kg)	(ft ²)	(m ²)	(CFM)	(m ³ /hr)
31~50	50	1.42	46	4.22	83	140
51~100	100	2.84	91	8.44	165	279
101~150	150	4.26	137	12.66	247	418
151~200	200	5.67	182	16.85	328	556
201~250	250	7.09	227	21.07	410	696
251~300	300	8.51	273	25.29	492	835
301~350	350	9.93	318	29.51	574	974
351~400	400	11.34	363	33.69	655	1112
401~450	450	12.76	409	37.91	737	1251
451~500	500	14.18	454	42.13	819	1391
501~550	550	15.60	499	46.35	901	1530
551~600	600	17.01	545	50.54	982	1668

TABLE 6

8 Condensate Drain Lines

The coil drain pan has a primary and a secondary drain with 3/4" NPT female connections. The connectors required are 3/4" NPT male, either PVC or metal pipe, and should be hand tightened to a torque of no more than 37 in-lbs. to prevent damage to the drain pan connection. An insertion depth of approximately 3/8" to 1/2" (3-5 turns) should be expected at this torque.

1. Ensure drain pan hole is not obstructed.
2. To prevent potential sweating and dripping on to finished space, it may be necessary to insulate the condensate drain line located inside the building. Use Armaflex® or similar material.

A secondary condensate drain connection has been provided for areas where the building codes require it. Pitch all drain lines a minimum of 1/4" per foot to provide free drainage. Provide required support to the drain line to prevent bowing.

If the secondary drain line is required, run the line separately from the primary drain and end it where condensate discharge can be easily seen.

To ensure proper drainage, the Δ between the front of the unit and the back of the unit must be $-\frac{1}{4}" \leq 0"$ - where $-\frac{1}{4}"$ is measured as the difference between the front and the back of the unit. The 0" conveys that the front and back of the unit are completely level.

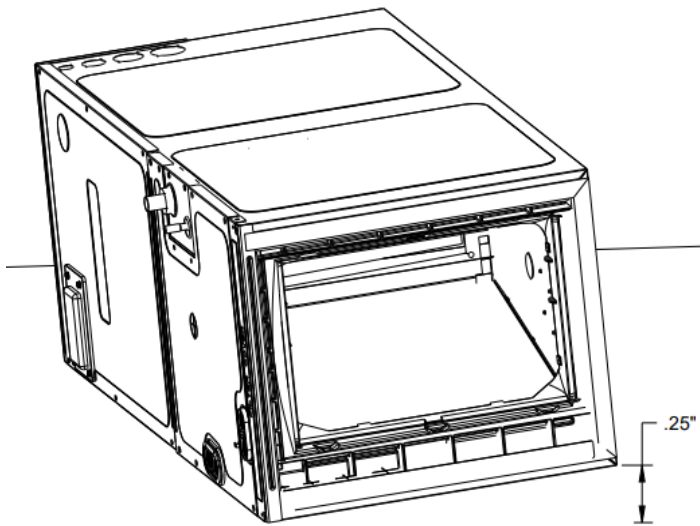


FIGURE 16

NOTE: Water coming from secondary line means the coil primary drain is plugged and needs immediate attention.



CAUTION

IF SECONDARY DRAIN IS NOT INSTALLED, THE SECONDARY ACCESS MUST BE PLUGGED.

Insulate drain lines located inside the building or above a finished living space to prevent sweating. Install a condensate trap to ensure proper drainage.

NOTE: When units are installed above ceilings, or in other locations where damage from condensate overflow may occur, it is **MANDATORY** to install a field fabricated auxiliary drain pan under the coil cabinet enclosure.

The installation must include a "P" style trap that is located as close as is practical to the evaporator coil. See Figure 20 for details of a typical condensate line "P" trap.

NOTE: Units operating in high static pressure applications may require a deeper field constructed "P" style trap than is shown in Figure 17 to allow proper drainage and prevent condensate overflow.

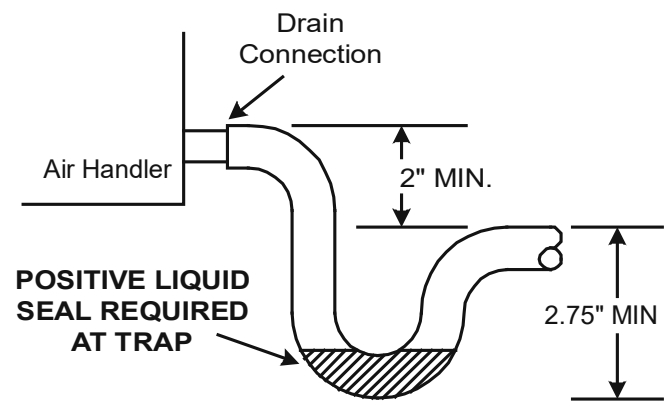


FIGURE 17

NOTE: Trapped lines are required by many local codes. In the absence of any prevailing local codes, please refer to the requirements listed in the Uniform Mechanical Building Code.

A drain trap in a draw-through application prevents air from being drawn back through the drain line during fan operation thus preventing condensate from draining, and if connected to a sewer line to prevent sewer gases from being drawn into the airstream during blower operation.

Use of a condensate removal pump is permitted when necessary. This condensate pump should have provisions for shutting off the control voltage should a blocked drain occur. A trap must be installed between the unit and the condensate pump.

IMPORTANT NOTE: The evaporator coil is fabricated with oils that may dissolve styrofoam and certain types of plastics. Therefore, a removal pump or float switch must not contain any of these materials.

Tip: Priming the "P" trap may avoid improper draining at the initial installation and at the beginning of the cooling season.

9 Ductwork

This air handler is designed for a complete supply and return ductwork system.



WARNING

ALL REQUIRED VENTILATION OPENINGS MUST BE CLEAR OF OBSTRUCTION AND ALL DUCTS CONNECTED TO THE APPLIANCE FREE OF POTENTIAL IGNITION SOURCES.



CAUTION

DO NOT OPERATE THIS PRODUCT WITHOUT ALL THE DUCTWORK ATTACHED.

To ensure correct system performance, the ductwork is to be sized to accommodate 350-450 CFM per ton of cooling with the static pressure not to exceed 0.5" in w.c. Refer to ACCA Manual D, Manual S and Manual RS for information on duct sizing and application. Flame retardant ductwork is to be used and sealed to the unit in a manner that will prevent leakage.

NOTE: A downflow application with electric heat must have an L-shaped sheet metal supply duct without any outlets or registers located directly below the heater.

9.1 RETURN DUCTWORK

Instructions for wiring to external zoning dampers and/or mechanical ventilation, if required to ensure that upon detection of a leak, the zoning dampers are driven fully open and additional mechanical ventilation is activated.

For appliances using R-32 REFRIGERANTS, connected via an air duct system to one or more rooms, the supply and return air shall be directly ducted to the space. Open areas such as false ceilings shall not be used as a return air duct.

For mechanical ventilation the air extraction opening from the room shall be located equal or below the refrigerant release point. For floor mounted units, it shall be as low as practicable. The air extraction openings shall be located in a sufficient distance from the air intake openings to prevent re-circulation to the space.

DO NOT LOCATE THE RETURN DUCTWORK IN AN AREA THAT CAN INTRODUCE TOXIC, OR OBJECTIONABLE FUMES/ODORS INTO THE DUCTWORK. The return ductwork is to be connected to the air handler bottom (upflow configuration).

9.2 SUPPLY DUCTWORK

Supply ductwork should be attached to the provided plenum rails on the air handler to avoid excessive condensation in the control box. If supply plenum encloses the entire top of the unit, it is recommended that the top plate of the air handler be insulated to prevent condensation. See Figure 18.

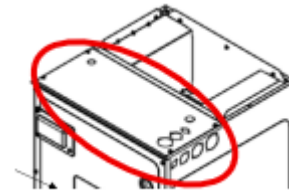


FIGURE 18

10 Return Air Filters

NOTE: For AMST24BU1300 only: If installing a filter, an external filter must be used when installing the unit in Horizontal Right, or if the side drain pan is not removed for Down Flow application. A filter will not fit on the internal filter rails in these applications.**

Each installation must include a return air filter. This filtering may be performed at the air handler using the factory filter rails or externally such as a return air filter grille. When using the factory filter rails, a nominal 16x20x1", 20x20x1" or 24x20x1" (actual dimension must be less than 23-1/2"x20") filter can be installed on a B, C and D cabinet respectively (the cabinet size is the seventh letter of the model number).

IMPORTANT NOTE: If appliance is equipped with Ultraviolet coil and air purifier, turn off the appliance before opening filter access door panel to change and install filter.

11 Electric Heat

Refer to the installation manual provided with the electric heat kit for the correct installation procedure. All electric heat must be field installed. Transformer must be detached during heater kit installation process. If installing this option, the ONLY heat kit that is permitted to be used is HKTS* series. Refer to the air handler unit's Serial and Rating plate or the HKTS* specification sheets to determine the heat kits compatible with a given air handler. No other accessory heat kit besides the HKTS* series may be installed in these air handlers.

The heating mode temperature rise is dependent upon the system airflow, the supply voltage, and the heat kit size (kW) selected. Use data provided in Tables 2, 3, and 4 to determine the temperature rise (°F).

NOTE: For installations not indicated above the following formula is to be used:

$$TR = (kW \times 3412) \times (\text{Voltage Correction}) / (1.08 \times \text{CFM})$$

Where: TR	=	Temperature Rise
kW	=	Heater Kit Actual kW
3412	=	Btu per kW
VC*	=	.96 (230 Supply Volts)
	=	.92 (220 Supply Volts)
	=	.87 (208 Supply Volts)
1.08	=	Constant
CFM	=	Measured Airflow

*VC (Voltage Correction)

NOTE: The Temperature Rise Tables can also be used to estimate the air handler airflow delivery. When using these tables for this purpose set the room thermostat to maximum heat and allow the system to reach steady state conditions. Insert two thermometers, one in the return air and one in the supply air. The temperature rise is the supply air temperature minus the room air temperature. Using the temperature rise calculated, CFM can be estimated from the TR formula above. See Service Manual for more information.

CFM	HEAT KIT NOMINAL kW							
	3	5	6	8	10	15	19/20	25
800	11	18	22	30	35			
1000	9	14	18	24	28	42		
1200	7	12	15	20	24	35	47	59
1400	6	10	13	17	20	30	40	51
1600	6	9	11	15	18	27	35	44
1800	5	8	10	13	16	24	31	39
2000	4	7	9	12	14	21	28	35

220/1/60 SUPPLY VOLTAGE - TEMP. RISE °F

TABLE 8

CFM	HEAT KIT NOMINAL kW							
	3	5	6	8	10	15	19/20	25
800	10	17	21	28	33			
1000	8	13	17	22	27	40		
1200	7	11	14	19	22	33	45	56
1400	6	10	12	16	19	29	38	48
1600	5	8	10	14	17	25	33	42
1800	5	7	9	12	15	22	30	37
2000	4	7	8	11	13	20	27	33

208/1/60 SUPPLY VOLTAGE - TEMP. RISE °F

TABLE 9

CFM	HEAT KIT NOMINAL kW							
	3	5	6	8	10	15	19/20	25
800	12	19	23	31	37			
1000	9	15	19	25	30	44		
1200	8	12	15	21	25	37	49	62
1400	7	11	13	18	21	32	42	53
1600	6	9	12	15	19	28	37	46
1800	5	8	10	14	16	25	33	41
2000	5	7	9	12	15	22	30	37

230/1/60 SUPPLY VOLTAGE - TEMP. RISE °F

TABLE 7

Model	HEATER KIT (kW)								
	3	5	6	8	10	15	19	20	25
AMST24BU13	715	715	715	715	850	850			
AMST30BU13	715	715	715	715	875	1050			
AMST36BU13	715	715	715	715	875	1050			
AMST36CU13	1170	1170	1170	1170	1170	1345	1345		
AMST42CU13	1170	1170	1170	1170	1170	1345	1345		
AMST48CU13	1170	1170	1170	1170	1170	1345	1345		
AMST60DU13	1590	1590	1590	1590	1590	1715		1715	1930

*AIRFLOWS SHOWN ARE TO BE CONSIDERED THE ABSOLUTE MINIMUM ALLOWABLE FOR THE AIR HANDLER AND HEAT KIT COMBINATION. THE MINIMUM AIRFLOW DOES NOT REPRESENT THE RECOMMENDED AIRFLOW BY THE MANUFACTURER. WHEN SELECTING A HEATER KIT, THE MINIMUM BLOWER SETTING (M.B.S.) OR SPEED TAP LISTED ON THE UNIT'S NAMEPLATE SHOULD BE FOLLOWED.

MINIMUM CFM REQUIRED FOR HEATER KITS

TABLE 10

12 Airflow Data

Model	Speed Tap	CFM @ STATIC PRESSURE (IN W.C.)								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
AMST24BU1300BA	T1	825	800	745	730	660	645	560	550	460
	T2	850	825	770	750	675	665	590	575	470
	T3	935	910	865	850	790	780	715	705	620
	T4	1045	1025	985	970	920	910	850	845	785
	T5	1100	1080	1035	1020	970	960	905	900	840
	T6	900	875	830	815	750	740	670	655	550
	T7	1030	1005	960	950	900	890	825	820	760
	T8	1215	1195	1155	1145	1105	1095	1045	1040	980
	T9	1215	1195	1155	1145	1105	1095	1045	1040	980
AMST30BU1300BA	T1	855	830	780	765	705	695	625	615	515
	T2	920	895	845	830	780	770	710	700	615
	T3	985	960	915	900	850	840	785	775	710
	T4	1070	1050	1005	995	945	935	890	885	830
	T5	1185	1165	1125	1115	1070	1060	1015	1010	960
	T6	985	960	915	900	850	840	785	775	710
	T7	1070	1050	1005	995	945	935	890	885	830
	T8	1185	1165	1125	1115	1070	1060	1015	1010	960
	T9	1185	1165	1125	1115	1070	1060	1015	1010	960
AMST36BU1300BA	T1	855	830	780	765	705	695	625	615	515
	T2	920	895	845	830	780	770	710	700	615
	T3	985	960	915	900	850	840	785	775	710
	T4	1070	1050	1005	995	945	935	890	885	830
	T5	1185	1165	1125	1115	1070	1060	1015	1010	960
	T6	985	960	915	900	850	840	785	775	710
	T7	1070	1050	1005	995	945	935	890	885	830
	T8	1185	1165	1125	1115	1070	1060	1015	1010	960
	T9	1185	1165	1125	1115	1070	1060	1015	1010	960
AMST36CU1300BA	T1	1070	1035	960	935	830	810	700	690	610
	T2	1165	1130	1055	1035	960	940	825	815	715
	T3	1285	1255	1190	1170	1100	1085	990	980	885
	T4	1430	1400	1345	1330	1260	1250	1175	1165	1065
	T5	1560	1530	1470	1455	1390	1380	1310	1300	1235
	T6	1430	1400	1345	1330	1260	1250	1175	1165	1065
	T7	1735	1710	1660	1645	1580	1570	1510	1500	1440
	T8	1830	1805	1755	1740	1685	1675	1605	1595	1525
	T9	1830	1805	1755	1740	1685	1675	1605	1595	1525
AMST42CU1300BA	T1	1165	1140	1085	1065	990	975	895	880	765
	T2	1230	1205	1150	1135	1070	1060	990	975	850
	T3	1410	1385	1335	1325	1280	1270	1195	1185	1115
	T4	1440	1415	1365	1355	1305	1295	1235	1225	1155
	T5	1495	1470	1425	1415	1365	1355	1305	1295	1220
	T6	1580	1550	1510	1490	1450	1420	1380	1330	1300
	T7	1410	1385	1335	1325	1280	1270	1195	1185	1115
	T8	1760	1730	1700	1670	1640	1610	1580	1550	1505
	T9	1760	1730	1700	1670	1640	1610	1580	1550	1505
AMST48CU1300BA	T1	1420	1390	1330	1310	1235	1220	1135	1125	1050
	T2	1465	1435	1375	1355	1280	1265	1165	1155	1065
	T3	1580	1550	1490	1470	1400	1390	1315	1305	1230
	T4	1635	1610	1560	1545	1480	1470	1400	1390	1315
	T5	1735	1710	1660	1640	1560	1550	1485	1475	1410
	T6	1420	1390	1330	1310	1235	1220	1135	1125	1050
	T7	1800	1775	1730	1715	1675	1635	1595	1550	1460
	T8	1820	1795	1750	1735	1680	1670	1605	1595	1525
	T9	1820	1795	1750	1735	1680	1670	1605	1595	1525
AMST60DU1300BA	T1	1215	1175	1095	1070	975	950	790	780	700
	T2, T6	1815	1785	1725	1710	1650	1640	1570	1560	1490
	T3, T7	1360	1325	1250	1230	1145	1125	1010	990	850
	T4, T8	2070	2045	1995	1980	1930	1920	1855	1845	1785
	T5, T9	1970	1945	1895	1880	1815	1805	1740	1730	1660

* Gray cells are for additional information

AMST AIRFLOW DATA (CFM)

TABLE 11

NOTE:

4. Airflow data indicated is at 230V without air filter in place.
5. Static on table includes static from media filter. Motor Speed Tap should be selected with media filter included in static value.
6. The chart is for information only. For satisfactory operation, external static pressure must not exceed value shown on rating plate.
7. The cooling/heat pump speed tap should be selected based on the AHRI rating. Otherwise, select a speed tap that provides a minimum 350 CFM per outdoor ton.
8. Use the CFM adjustment factors of 0.98 for horizontal left and 0.96 for horizontal right & downflow orientations.
9. When applying a humidistat (normally closed), refer to the installation and operating instructions. The humidistat can adjust the cooling airflow to 85%.
10. Airflow test data is proved to satisfy minimum room area (Amin) and minimum airflow at mitigation mode (Qmin). See Table 6.
11. For installations with Heater Kits, the speed tap airflow selection must exceed the minimum airflow based on minimum CFM required for heater kit.

It is the installer's responsibility to verify that actuation of mitigation procedures are operational with minimum required airflow according to the minimum mitigation airflow chart (Refer to table 6).

13 ALTITUDE ADJUSTMENT FACTOR TO CALCULATE MINIMUM ROOM AREA

The Indoor equipment mitigation requirements are calculated at sea level. For higher altitudes adjust the minimum room area specified on the Serial Plate by the corresponding altitude adjustment factor shown below. This table is provided as a reference. Adjusted room area (Amin adj) is the product of the minimum air conditioned room area specified in the serial plate and the adjustment factor AF, as shown in below formula

$A_{min\ adj} = A_{min} \text{ (Minimum air conditioned room area from table 6)} * AF$

Height in meters	Height in feet	Altitude Adjustment Factor (AF)
At sea level	At sea level	1.00
1~200	1~660	1.02
200~400	660~1320	1.03
400~600	1320~1970	1.05
600~800	1970~2630	1.07
800~1000	2630~3290	1.09
1000~1200	3290~3940	1.11
1200~1400	3940~4600	1.13
1400~1600	4600~5250	1.15
1600~1800	5250~5910	1.17
1800~2000	5910~6570	1.19
2000~2200	6570~7220	1.21
2200~2400	7220~7880	1.24
2400~2600	7880~8540	1.26
2600~2800	8540~9190	1.29
2800~3000	9190~9850	1.31
3000~3200	9850~10500	1.34

TABLE 12

NOTE: The altitude adjustment factor (AF) for any altitudes below sea level will be 1.00

14 Electrical and Control Wiring

IMPORTANT: The appliance shall be installed in accordance with national wiring regulations such as the National Electric Code (NEC)/American National Standards Institute (ANSI)/National Fire Protection Association (NFPA) 70 and local/state codes. In Canada, electrical grounding is to be in accordance with the Canadian Electric Code (CSA) C22.1. All routing of electrical wiring must be made through provided electrical knockouts. Do not cut, puncture or alter the cabinet for electrical wiring.

WARNING

HIGH VOLTAGE

DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

WARNING

HIGH VOLTAGE!

TO AVOID PROPERTY DAMAGE, PERSONAL INJURY OR DEATH DUE TO ELECTRICAL SHOCK, THIS UNIT MUST HAVE AN UNINTERRUPTED, UNBROKEN ELECTRICAL GROUND. THE ELECTRICAL GROUND CIRCUIT MAY CONSIST OF AN APPROPRIATELY SIZED ELECTRICAL WIRE CONNECTING THE GROUND LUG IN THE UNIT CONTROL BOX TO THE BUILDING ELECTRICAL SERVICE PANEL. OTHER METHODS OF GROUNDING ARE PERMITTED IF PERFORMED IN ACCORDANCE WITH THE NATIONAL ELECTRIC CODE (NEC)/AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) 70 AND LOCAL/STATE CODES. IN CANADA, ELECTRICAL GROUNDING IS TO BE IN ACCORDANCE WITH THE CANADIAN ELECTRIC CODE (CSA) C22.1.

14.1 BUILDING ELECTRICAL SERVICE INSPECTION

DO NOT OPERATE THIS UNIT ON A THREE-PHASE POWER SUPPLY, ONLY A THREE-PHASE HEAT KIT CAN UTILIZE THREE-PHASE POWER. Measure the power supply to the unit. The supply voltage must be measured and be in agreement with the unit nameplate power requirements and within the range shown.

14.2 WIRE SIZING

Wire size is important to the operation of your equipment. Use the following check list when selecting the appropriate wire size for your unit.

- Wire used must carry the Minimum Circuit Ampacity (MCA) listed on the unit's Series and Rating Plate.
- Refer to the NEC (USA) or CSA (Canada) for wire sizing. The unit MCA for the air handler and the optional electric heat kit can be found on the unit Series and Rating Plate.
- Wire must be sized to allow no more than a 2% voltage drop from the building breaker/fuse panel to the unit.
- Wires with different insulation temperature rating have varying ampacities - be sure to check the temperature rating used.

Refer to the latest edition of the National Electric Code or in Canada the Canadian Electric Code when determining the correct wire size.

14.3 MAXIMUM OVERCURRENT PROTECTION (MOP)

Every installation must include an NEC (USA) or CEC (Canada) approved overcurrent protection device. Also, check with local or state codes for any special regional requirements.

Protection can be in the form of fusing or HACR style circuit breakers. The Serial and Rating Plate provides the maximum overcurrent device permissible.

NOTE: Fuses or circuit breakers are to be sized larger than the equipment MCA but not to exceed the MOP, values are marked on the equipment nameplate.

Nominal Input	Minimum Voltage	Maximum Voltage
208/230 VAC	197	253

**ELECTRICAL VOLTAGE
TABLE 13**

14.4 WIRING

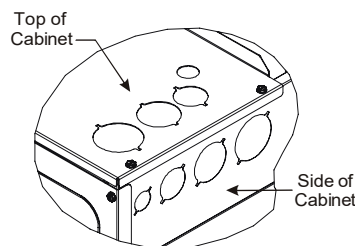
Check that wiring will not be subject to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects. The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans

14.5 ELECTRICAL CONNECTIONS – SUPPLY VOLTAGE

 CAUTION
FIRE HAZARD! TO AVOID THE RISK OF PROPERTY DAMAGE, PERSONAL INJURY OR FIRE, USE ONLY COPPER CONDUCTORS.

IMPORTANT NOTE: USE COPPER CONDUCTORS ONLY FROM DISCONNECT OR ELECTRICAL PANEL TO THE AIR HANDLER.

Knockouts are provided on the air handler top panel and sides of the cabinet to allow for the entry of the supply voltage conductors, as shown in Figure 19. Separate knockouts must be used for two circuit heat kits. Unless Single Point Wiring Kit (SPW-01) is used. If the knockouts on the cabinet sides are used for electrical conduit, an adapter ring must be used in order to meet UL-60335 safety requirements. An NEC or CEC approved strain relief is to be used at this entry point. Some codes/municipalities require the supply wire to be enclosed in conduit. Consult your local codes.



**KNOCK-OUT FOR ELECTRICAL CONNECTIONS
FIGURE 19**

14.5.1 AIR HANDLER ONLY (NON-HEAT KIT MODELS)

IMPORTANT: Installation of Air Handler must follow any local codes/regulations. The manufacture recommends that when a heater kit is not installed, a field supplied disconnect switch or breaker should be installed in the electrical circuit that will allow power to be shut-off for service or maintenance.

The building supply connects to the stripped black and red wires contained in the air handler electrical compartment cavity. A ground screw is also contained in this area. Attach the Supply wires to the air handler conductors as shown in the unit wiring diagram using appropriately sized solderless connectors or other NEC or CEC approved means.

14.5.2 AIR HANDLER - NON-CIRCUIT BREAKER HEAT KITS

A terminal block is provided with the HKTS* kit to attach the power supply and air handler connections. Follow the HKTS* Installation Manual and wiring diagram for complete wiring details.

14.5.3 AIR HANDLER WITH CIRCUIT BREAKER HEAT KIT

The air handler has soft plastic cover which will only allow to access the heater kit circuit breaker. It does not allow to install or replace the circuit breaker. The circuit breakers have lugs for power supply connection. See the HKTS* Installation Instructions for further details.

14.6 LOW VOLTAGE CONNECTIONS

Several combinations of low voltage schemes are possible, depending on the presence of a heat kit and whether the heat kit is single-stage or multi-stage, whether the outdoor section is an air conditioner or heat pump, and whether the outdoor section is single-stage or two-stage. The 24V-control voltage connects the air handler to the room thermostat and condenser. Low voltage wiring must be copper conductors. A minimum of 18AWG must be used for installations up to 100 feet. Low voltage wiring must be connected through the top of the cabinet or either side. See the "Thermostat Wiring" section of this manual for typical low voltage wiring connections.

14.6.1 THERMOSTATS

Second-stage heat can be accomplished by a multi-stage heating thermostat or the addition of an outdoor thermostat as shown in wiring schematics. Follow the thermostat manufacturer's instructions for installation.

14.7 SPEED TAP ADJUSTMENT

AMST**U1300** air handlers feature 9-speed energy efficient blower motors. The motors run at a constant torque with very low power consumption. Changing the 24 VAC leads to the desired speed tap on the PCB will adjust provided CFM (speed tap adjustment of the AMST60DU1300** is not supported).

The AMST blower motor speeds are programmed to deliver adequate airflow at rated external static pressure and with a 60 second off time delay. For details, refer to the specification sheet applicable to your model.

14.7.1 SELECTING SPEED TAPS 1-9 (NOT APPLICABLE TO AMST60DU1300**)

When selecting Speed Taps 1-9, follow the program tap table instruction to adjust the dip switch on the PCB. See table 14. (Also shown on the Wiring Diagram for 1.5T - 4T.)

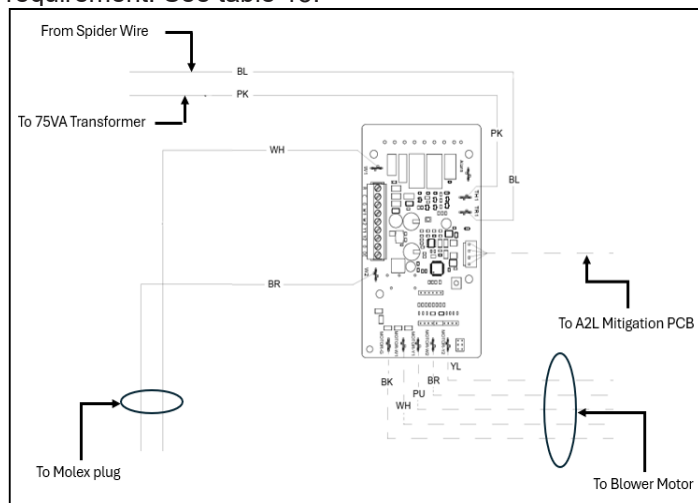
NOTE: Factory default program tap is 8.

PROGRAM TAP	9 TAP MODE				
	1	2	3	4	5
1	ON	OFF	OFF	OFF	OFF
2	OFF	ON	OFF	OFF	OFF
3	OFF	X	ON	OFF	OFF
4	OFF	X	X	ON	OFF
5	OFF	X	X	X	ON
6	ON	ON	OFF	OFF	OFF
7	ON	X	ON	OFF	OFF
8	ON	X	X	ON	OFF
9	ON	X	X	X	ON

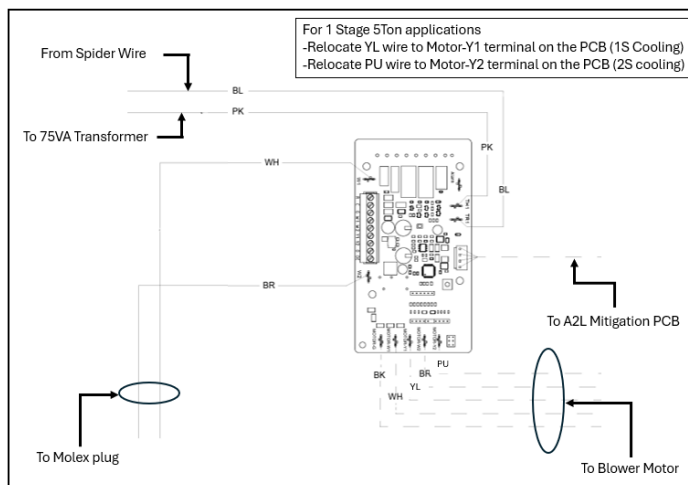
TABLE 14

14.7.2 AMST60DU1300** (ONLY)

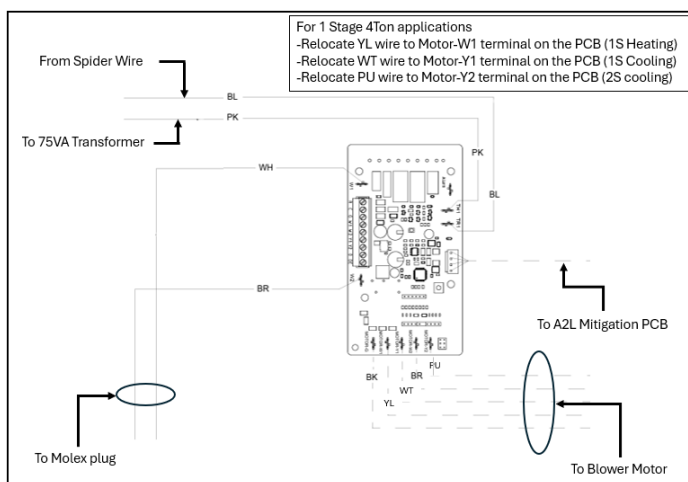
Wiring is setup for 2-Stage compressor speed application as default from factory Figure 20A. For 1-Stage application refer to Figure 20B or 20C depending on airflow requirement. See table 15.



DEFAULT 2-STAGE APPLICATIONS WIRING
FIGURE 20A



1-STAGE APPLICATIONS WIRING
FIGURE 20B



1-STAGE APPLICATIONS WIRING
FIGURE 20C

	Motor G	Motor W1	Motor Y1	Motor W2	Motor Y2
2-Stage Applications 5Ton	BK	WH	PU	BR	YL
1-Stage Applications 5Ton	BK	WH	YL	BR	PU
1-Stage Applications 4Ton	BK	YL	WT	BR	PU

TABLE 15

14.7.3 Accessory installations

For any and all accessories which may become a potential ignition source, they must be installed to our accessory control board. Ensure that any additional wiring from the Air Handler is done so as to protect from damage and wear or from any joints which may need to be brazed or disconnected for service. If a zone control system is installed in the ductwork attached to this system, the zone controller must be powered through a accessory control kit to ensure that the zoning dampers open during mitigation mode. Refer to the product specification sheet for the zoning and accessory control kit part number.

15 Air Handler Low Voltage Connections

15.1 HEATER KITS AND ECM MOTOR

The following composite wiring diagrams detail various configurations in which the air handlers can be used. Examples include single-stage cooling and heat pump with single or two-stage electric heating. All these configurations can be applied with convenient connections to outdoor thermostat applications.

The following sections are detailed:

- Single-Stage Cooling
- Heat Pump

Each diagram details the connections between room thermostat and the air handlers, and the connections between the air handlers and the Condensing Unit (or Heat Pump) with optional connections to Outdoor Thermostats. For each configuration, refer to the explanation of the proper jumper(s) to remove for the corresponding blower speed that will result in the programmed fixed speed ECM motor.

IMPORTANT: When matching the AMST air handlers to a single speed cooling or heat pump unit, remember to connect “Y” from the thermostat to the “Y2” on the low voltage terminal board.

Any equivalent thermostat can be used in place of the manufacturer’s thermostat part number.

15.2 REFRIGERATION DETECTION SYSTEM (RDS)

Function: Refrigerant Detection System (RDS) is installed in this equipment to detect any R32 leakage in the coil and take action to mitigate any risk of ignition/ fire.

Operation: When Refrigerant is detected, the RDS shall send signals for the unit to perform these actions:

1. Turn off Heating and cooling operation
2. Turn on the blower for air circulation
3. Switch off electric heater

The mitigation mode takes about 5 minutes to complete after which it will resume normal operation. The minimum mitigation airflow chart provides the required airflow and the minimum room area (In this case, the room area refers to the total conditioned airspace. Refer to table 11 and table 6 to disperse the refrigerant without causing a potential fire hazard in presence of ignition sources).

Servicing: Before servicing identify the Mode of operation of the system by reading the LED flashing pattern on the PCB which can be seen through the round glass view on top access panel and matching the LED flashing pattern with mode of operation in A2L PCB fault code label which is attached on the front side of the blower access panel (Figure 21). After identifying the mode of operation take recommended actions as specified in Table 17.

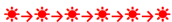
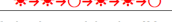
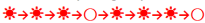
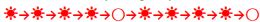
Serial Plate Information: The installer must write the system charge used on installation on the serial plate under the maximum possible refrigerant charge field. Consult table 6 to determine minimum air conditioned room area required for the system charge used. If installation is not at sea level, consult section 13 to determine how to adjust the minimum air conditioned room area. Write the minimum air conditioned room area on the serial plate under the minimum room area field. Use a durable marking instrument to write the values on the serial plate.

The installers are responsible for verifying the actuation of mitigation procedures are operational with minimum required airflow for mitigation (refer to table 11 for airflow data).

Instruction to replace A2L PCB: Take off the blower access panel, disconnect the PCB harness and R32 sensor wire connected to the PCB, detach the PCB from the 4 plastic standoffs, install new PCB on 4 plastic standoffs which is installed on metal bracket, re-connect the PCB harness and R32 sensor wire to the new PCB per wiring instruction as attached to the equipment, reassemble the blower access panel to the unit.

Instruction to replace R32 sensor: Take off the blower access panel and coil access panel, take off the drain port gasket on the drain pan in front of the sensor bracket, disconnect the R32 sensor wire from the A2L PCB, take off sensor bracket assembly from the drain pan, remove plastic push pins and R32 sensor off the sensor bracket, install new R32 sensor and plastic push pins to sensor bracket, re-install A2L sensor bracket assembly to the drain port correctly as shown in the figures 5, 6, 7 and 8. The “FRONT” print on sensor bracket should be facing away from the equipment, place gaskets back to the drain ports correctly. The “FRONT” print on the gaskets should be in the front facing away from the equipment, reassemble the blower access panel and coil access panel to the unit.

IMPORTANT NOTE: The R32 Sensor shall only be replaced with sensors specified by the appliance manufacturing. (Sensor replacement part can be determined through the homeowner support line provided earlier in this document.)

LED STATUS CODES		
RED LED Flashing Pattern	GREEN LED Flashing Pattern	MODE
OFF 	Slow LED flashing pattern (2 seconds on and 2 seconds off) 	Normal Operation
OFF 	Fast LED Flashing Pattern 	Manual System Test (Push button 2 times to start. Push button 1 time to stop)
Fast LED Flashing Pattern 	Slow LED flashing pattern (2 seconds on and 2 seconds off) 	R32 Leak Alarm (Control detects active R32 Leak)
LED on continuously 	Slow LED flashing pattern (2 seconds on and 2 seconds off) 	Delay Mode (system will return to Normal Operation in 5 minutes)
LED will flash 2 times and then be off for 5 seconds before repeating the pattern 	OFF 	Control Board Internal Fault (Replace the control board)
LED will flash 3 times and then be off for 5 seconds before repeating the pattern 	Slow LED flashing pattern (2 seconds on and 2 seconds off) 	R32 Sensor Communication Fault (Check R32 Sensor connection and Cycle Power - Replace R32 sensor if error continues)
LED will flash 4 times and then be off for 5 seconds before repeating the pattern 	Slow LED flashing pattern (2 seconds on and 2 seconds off) 	R32 Sensor Fault (Check R32 Sensor connection and Cycle Power - Replace R32 sensor if error continues)
LEAK DETECTION SYSTEM INSTALLED. UNIT MUST BE POWERED EXCEPT FOR SERVICE. *IF R32 LEAK ALARM IS OBSERVED, DO NOT OPEN THE UNIT OR TURN IT OFF		

A2L PCB FAULT CODE

FIGURE 21

RECOMMENDED ACTION FOR PCB LED FLASHING CODES

	service indicator; read technical manual
	operator's manual; operating instructions
	warning; flammable material
	UN GHS flame symbol

MARKING SYMBOL TABLE
TABLE 16

16 Achieving 1.4% Low Leakage Rate

Ensure all the gaskets remain intact on all surfaces as shipped with the unit. These surfaces are areas between the upper tie plate and blower access panel, blower access and coil access panels, and between the coil access and filter access panels. Ensure upon installation, that the plastic breaker cover is sitting flush on the blower access panel and all access panels are flush with each other and the cabinet. With these requirements satisfied, the unit achieves less than 1.4% airflow leakage when tested in accordance with ASHRAE Standard 193.

17 Start-Up Procedure



WARNING



HIGH VOLTAGE
DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

- Prior to start-up, ensure that all electrical wires are properly sized and all connections are properly tightened.
- All panels must be in place and secured. For Air Tight application, gasket must be positioned at prescribed locations to achieve 1.4% leakage.
- Tubing must be leak free.
- Condensate line must be trapped and pitched to allow for drainage.
- Low voltage wiring is properly connected.
- Auxiliary drain is installed when necessary and pitched to allow for drainage.
- Unit is protected from vehicular or other physical damage.
- Return air is not obtained from, nor are there any return air duct joints that are unsealed in, areas where there may be objectionable odors, flammable vapors or products of combustion such as carbon monoxide (CO), which may cause serious personal injury or death.

18 Regular Maintenance

18.1 CIRCULATING AIR FILTERS

The only item required to be maintained on a regular basis by the user is the circulating air filter(s). Filter should be cleaned or replaced regularly, typically once per month. A certified service technician must perform all other services.

IMPORTANT NOTE: If thumb screws are used to access the filter, ensure the washer installed on the screw behind the access panel remains in place after re-installation.

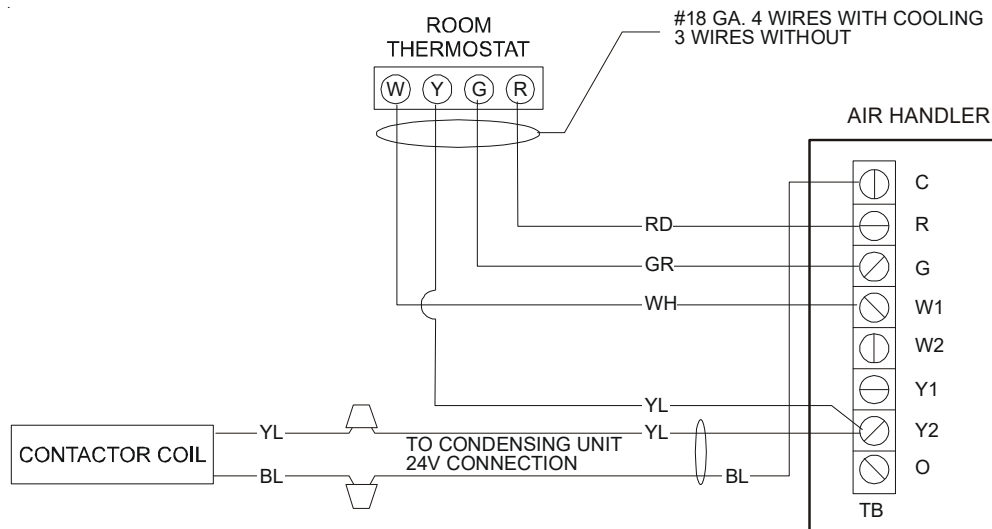
"LED STATUS CODES ● (LED ON) ○ (LED OFF) * (LED FAST FLASH)				
RED LED Flashing Pattern	GREEN LED Flashing Pattern	MODE	Definition	Recommended actions
"OFF ○	"Slow LED flashing pattern (2 seconds on and 2 seconds off) ●→○→●→○→●→○	Normal Operation	No faults to report	No action needed. Control system is working properly.
"OFF ○	"Fast LED Flashing Pattern *→*→*→*→*→*	System Test Mode	Manual test run by contractor to simulate R32 Leak Alarm (test will last for 5 minutes max)	This is only a test: No troubleshooting actions needed To Enter system verification test mode, press the button on the control board 2 times within 5 seconds. The control board will enter a simulated R32 Leak Alarm state and remain in that mode for 5 minutes. After 5 minutes, the control board will return to Normal Operation automatically. If the contractor wants to end the test early they need to press the button one time.
"Fast LED Flashing Pattern *→*→*→*→*→*	"Slow LED flashing pattern (2 seconds on and 2 seconds off) ●→○→●→○→●→○	R32 Leak Alarm	R32 refrigerant is currently being detected	The control board and sensor are working properly. Identify where the leak is coming from and address the leak.
"LED on continuously ●	"Slow LED flashing pattern (2 seconds on and 2 seconds off) ●→○→●→○→●→○	Delay Mode	After an R32 leak or Alarm has cleared, the unit will remain in R32 mitigation Delay Mode for 5 minutes before returning to normal operation	After any alarm or fault has cleared, the system will remain in R32 mitigation Delay Mode for 5 minutes and then return to normal operation. Wait until the control board returns to normal operation (LED Flashing Pattern will be SLOW - 2 seconds on and 2 seconds off). Check HVAC performance (Cooling and Heating Modes). Check system pressures and lines for any leaks. Re-Check HVAC performance after addressing any issues.
"LED will flash 2 times and then be off for 5 seconds before repeating the pattern *→*→○→*→*→○	"OFF ○	Control Board Internal Fault	Control board has detected an issue with the R32 detection system	The control board has detected an internal failure. Replace the control board. Note: if the control board was already replaced, and is still reporting a 2 flash pattern, Replace the sensor.
"LED will flash 3 times and then be off for 5 seconds before repeating the pattern *→*→*→○→*→*→○	"Slow LED flashing pattern (2 seconds on and 2 seconds off) ●→○→●→○→●→○	R32 Sensor Communication Fault	Control board does not have communications with the R32 sensor	1. Unplug and plug the R32 sensor back in. Cycle power to the system and wait 2 minutes. 2. If the control board is in Normal Operation (Slow Flash Pattern) or Delay mode (LED continuously on), there is no more issue. If not, continue with diagnostics 3. If the control board still displays ""R32 Sensor Communication Fault"" (3 flash pattern), Replace the sensor. Note: if the sensor was already replaced, and is still reports a 3 flash pattern, Replace the control board.
"LED will flash 4 times and then be off for 5 seconds before repeating the pattern *→*→*→*→○→*→*→*→○	"Slow LED flashing pattern (2 seconds on and 2 seconds off) ●→○→●→○→●→○	R32 Sensor Fault	R32 Sensor has reported an internal issue	1. Unplug and plug the R32 sensor back in. Cycle power to the system and wait 2 minutes. 2. If the control board is in Normal Operation (Slow Flash Pattern) or Delay mode (LED continuously on), there is no more issue. If not, continue with diagnostics 3. If the control board still displays ""R32 Sensor Fault"" (4 flash pattern), Replace the sensor.

TABLE 17

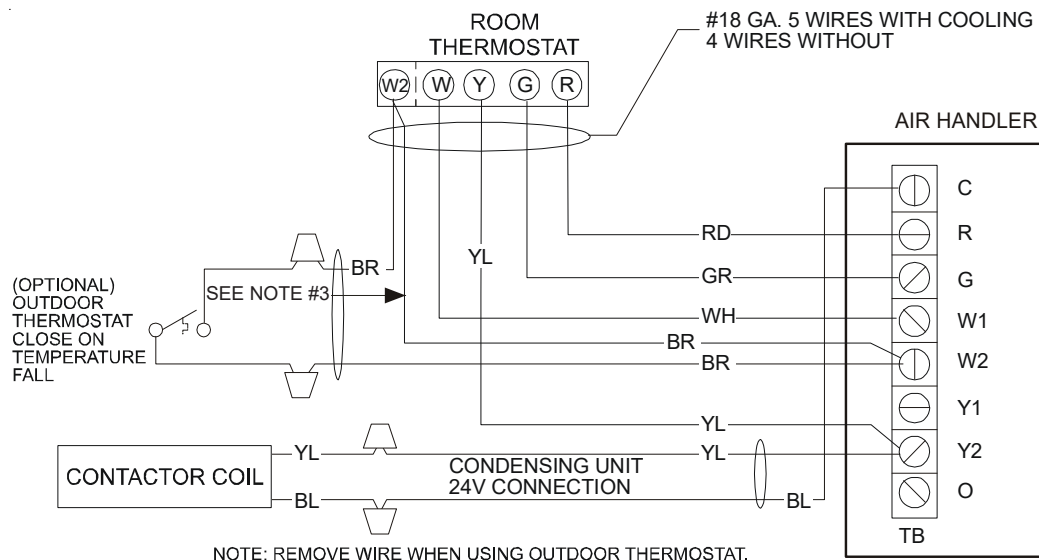
19 Wiring Diagrams

WARNING

HIGH VOLTAGE!
DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT.
MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY
CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



COOLING UNIT WITH OPTIONAL HEAT KITS OF 10 kW AND BELOW



**COOLING UNIT WITH OPTIONAL HEAT KITS OF 15 kW AND ABOVE
AND ROOM THERMOSTAT WITH TWO STAGES OF HEAT**

NOTES:

- 1) OUTDOOR THERMOSTAT (OT-1) SHOULD BE THE FIRST TO CLOSE AND THE LAST TO OPEN.
- 2) JUMPER W1 AND W2 TOGETHER IF OT-2 IS NOT USED.
- 3) REMOVE WIRE WHEN USING OUTDOOR THERMOSTAT.

NOMENCLATURE:

OT - OUTDOOR THERMOSTAT (OPTIONAL)
EHR - EMERGENCY HEAT RELAY (OPTIONAL)

COLOR CODES

RD - RED
YL - YELLOW
BL - BLUE
BR - BROWN
OR - ORANGE
WH - WHITE

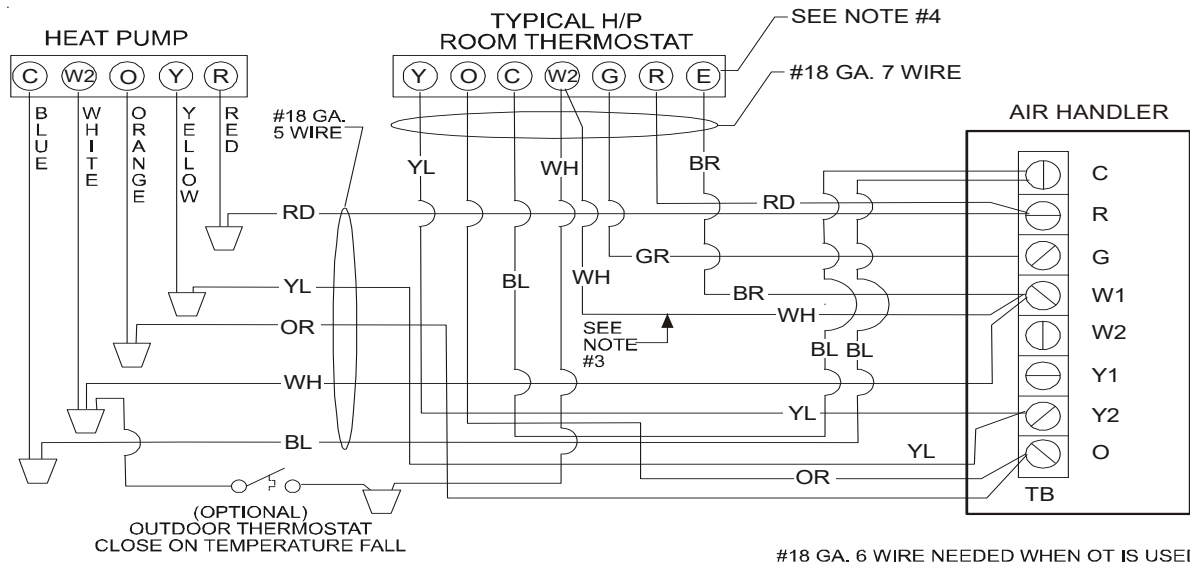
Wiring is subject to change. Always refer to the wiring diagram on the unit for the most up-to-date wiring.



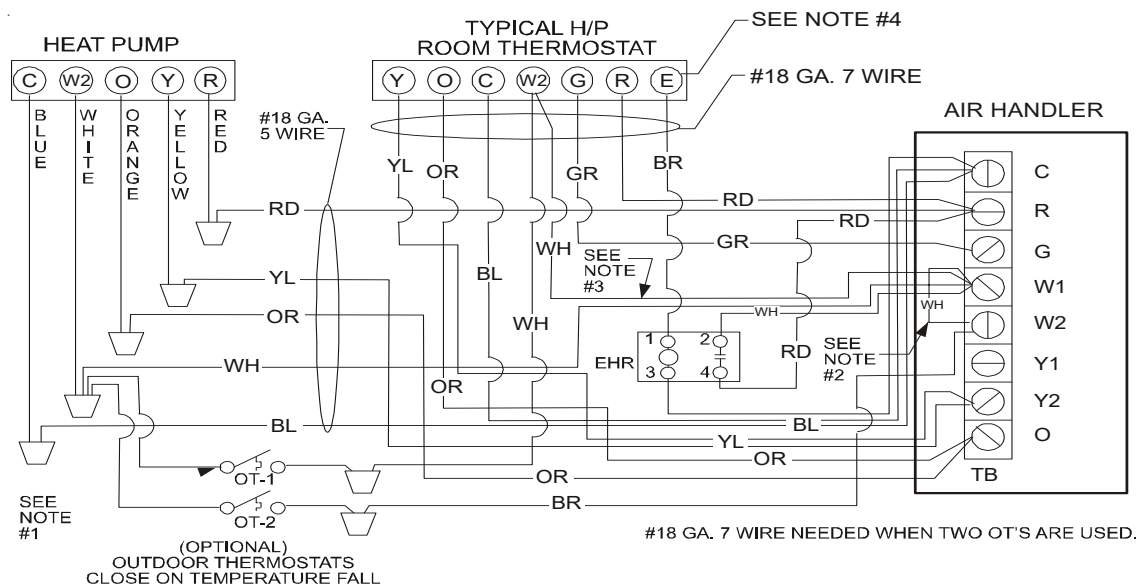
WARNING

HIGH VOLTAGE!

DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT.
MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY
CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



HEAT PUMP UNIT WITH OPTIONAL HEAT KITS OF 10 kW AND BELOW



HEAT PUMP UNIT WITH OPTIONAL HEAT KITS OF 15 kW AND ABOVE

NOTES:

- 1) OUTDOOR THERMOSTAT (OT-1) SHOULD BE THE FIRST TO CLOSE AND THE LAST TO OPEN.
- 2) JUMPER W1 AND W2 TOGETHER IF OT-2 IS NOT USED.
- 3) REMOVE WIRE WHEN USING OUTDOOR THERMOSTAT.
- 4) E TERMINAL MAY NOT BE APPLICABLE FOR SOME THERMOSTATS.

NOMENCLATURE:

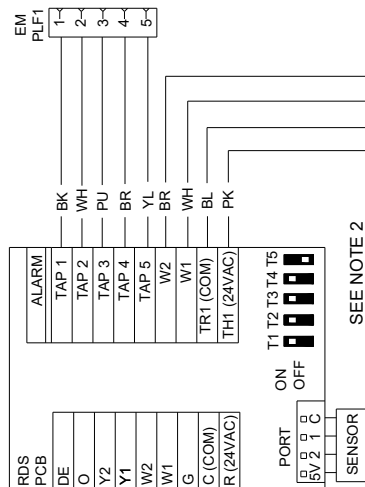
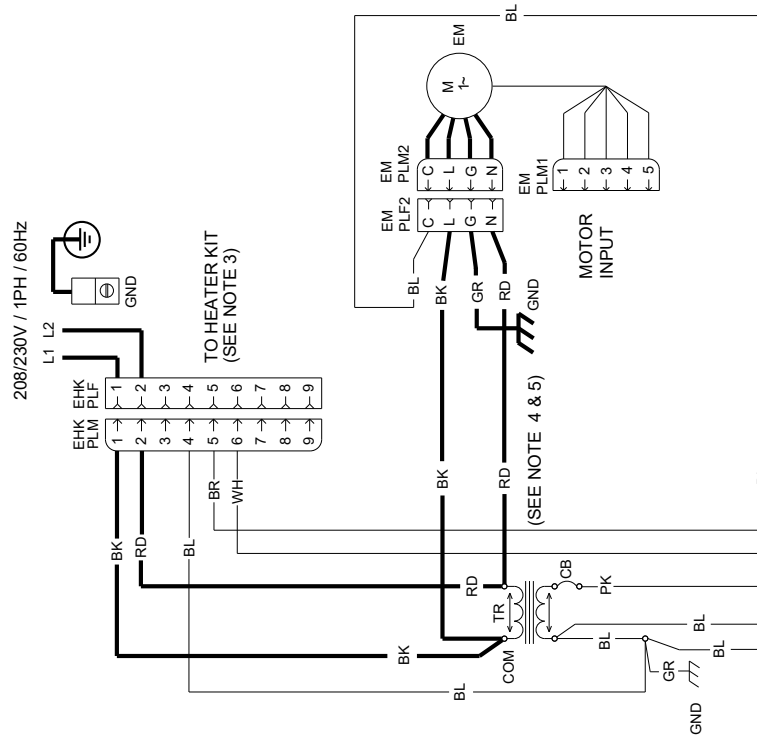
OT - OUTDOOR THERMOSTAT (OPTIONAL)
EHR - EMERGENCY HEAT RELAY (OPTIONAL)

COLOR CODES

RD - RED
YL - YELLOW
BL - BLUE
BR - BROWN
OR - ORANGE
WH - WHITE

Wiring is subject to change. Always refer to the wiring diagram on the unit for the most up-to-date wiring.

PROGRAM TAP	9 TAP MODE				
	1	2	3	4	5
1	ON	OFF	OFF	OFF	OFF
2	OFF	ON	OFF	OFF	OFF
3	OFF	X	ON	OFF	OFF
4	OFF	X	X	ON	OFF
5	OFF	X	X	X	ON
6	ON	ON	OFF	OFF	OFF
7	ON	X	ON	OFF	OFF
8	ON	X	X	ON	OFF
9	ON	X	X	X	ON



SEE NOTE 1

SEE NOTE 2

NOTES:

1. REPLACEMENT WIRE MUST BE SAME SIZE AND TYPE OF INSULATION AS ORIGINAL (AT LEAST 105° C). USE COPPER CONDUCTORS ONLY. USE N.E.C. CLASS 2 WIRE FOR ALL LOW VOLTAGE FIELD CONNECTIONS.
2. IF BLOWER MOTOR TAP NEEDS TO BE CHANGED, SEE INSTALLATION INSTRUCTIONS FOR FURTHER DETAILS ON ADJUSTING MOTOR.
3. DISCARD CONNECTOR EHK PLG WHEN INSTALLING OPTIONAL HEAT KIT.
4. RED WIRES CONNECTED TO 240V TAP AND BLACK WIRES CONNECTED TO COM TAP AT THE FACTORY. FOR 208V POWER SUPPLY, MOVE BOTH RED WIRES FROM 240V TAP TO THE 208V TAP.
5. LOW VOLTAGE TRANSFORMER RATED 24VAC OUTPUT 75VA @ 3.125 AMP.

COMPONENT LEGEND

CB	CIRCUIT BREAKER
EHK	ELECTRIC HEATER KIT
EM	EVAPORATOR MOTOR
GND	GROUND
PCB	PRINTED CIRCUIT BOARD
PLF	FEMALE PLUG
PLM	MALE PLUG
PRDS	REFRIGERANT DETECTION SYSTEM
TR	TRANSFORMER
COM	COMMON

FACTORY WIRING

 HIGH VOLTAGE
 OPTIONAL HIGH VOLTAGE
 LOW VOLTAGE
 OPTIONAL LOW VOLTAGE
 CHASSIS GROUND

FIELD WIRING

 HIGH VOLTAGE
 LOW VOLTAGE
 EARTH GROUND
 READ THE INSTRUCTIONS
 CLASS III



0140A20018-A



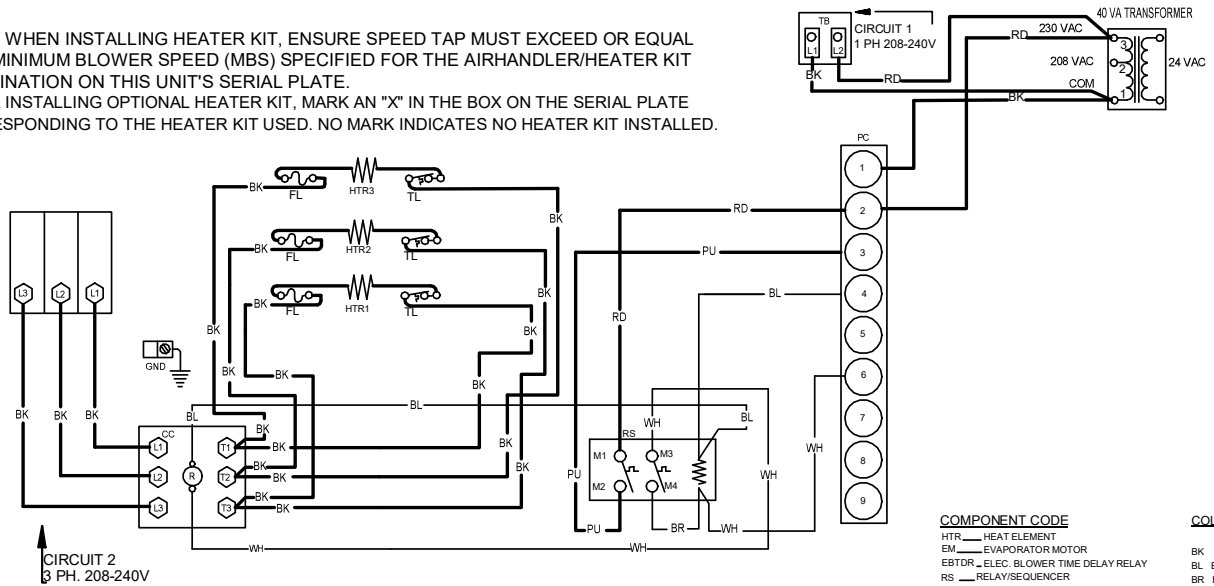
WARNING

HIGH VOLTAGE!

DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT.
MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY
CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



NOTE: WHEN INSTALLING HEATER KIT, ENSURE SPEED TAP MUST EXCEED OR EQUAL
THE MINIMUM BLOWER SPEED (MBS) SPECIFIED FOR THE AIRHANDLER/HEATER KIT
COMBINATION ON THIS UNIT'S SERIAL PLATE.
AFTER INSTALLING OPTIONAL HEATER KIT, MARK AN "X" IN THE BOX ON THE SERIAL PLATE
CORRESPONDING TO THE HEATER KIT USED. NO MARK INDICATES NO HEATER KIT INSTALLED.



NOTE:-
THIS LABEL MUST BE
ATTACHED ON THE
AIRHANDLER IN A PROMINENT
LOCATION

COMPONENT CODE

HTR — HEAT ELEMENT
EM — EVAPORATOR MOTOR
EBTDR — ELEC. BLOWER TIME DELAY RELAY
RS — RELAY/SEQUENCER
RC — RUN CAPACITOR
TL — THERMAL LIMIT
TR — TRANSFORMER
TB — TERMINAL BLOCK
CC — CONTACTOR
PC — 9-PIN CONNECTOR
SR — STRAIN RELIEF
FL — FUSE LINK
CB — CIRCUIT BREAKER

COLOR CODE

BK — BLACK
BL — BLUE
BR — BROWN
GR — GREEN
OR — ORANGE
PK — PINK
PU — PURPLE
RD — RED
WH — WHITE
YL — YELLOW

WIRING CODE

HIGH VOLTAGE ————
LOW VOLTAGE ————



0140M00273

AIR HANDLER

AIR HANDLER HOMEOWNER'S ROUTINE MAINTENANCE RECOMMENDATIONS

Replace or Clean Filter

IMPORTANT NOTE: Never operate unit without a filter installed as dust and lint will build up on internal parts resulting in loss of efficiency, equipment damage and possible fire.

An indoor air filter must be used with your comfort system. A properly maintained filter will keep the indoor coil of your comfort system clean. A dirty coil could cause poor operation and/or severe equipment damage.

Your air filter or filters could be located in your Air handler, in a blower unit, or in "filter grilles" in your ceiling or walls. The installer of your air conditioner or heat pump can tell you where your filter(s) are, and how to clean or replace them.

Check your filter(s) at least once a month. When they are dirty, replace or clean as required. Disposable type filters should be replaced. Reusable type filters may be cleaned.

You may want to ask your dealer about high efficiency filters. High efficiency filters are available in both electronic and non-electronic types. These filters can do a better job of catching small airborne particles.

Ultraviolet Coil and Air Purifiers

The listed Clean Comfort brand UV coil purifiers models are the only models approved to be used with this unit. They can be purchased from the distributors. The use of the accessory control kit is optional for the installation of these models. Maximum UV lamp diameter to be used is 1.375" to reduce the possibility of air leak.

The power source must be installed external of the unit. Ensure that the wiring is routed through the access panel knockout with the grommet included. A magnetic door switch must be installed to avoid an accidental ultraviolet exposure during service.

Refer to UV coil purifiers product specification and installation manual for additional details. See table 18 for models part numbers.

UV-C Part Numbers	
Models	Lamp
UC18S15-24	UCP-16013
UC18S15-24B	UCP-16012

TABLE 18

UVC light conversion kit is required to install with the UVC light. For installation refer to the UVC drain pan kit installation manual IO -7023 for additional details. See table 19 for model and kit assignment.

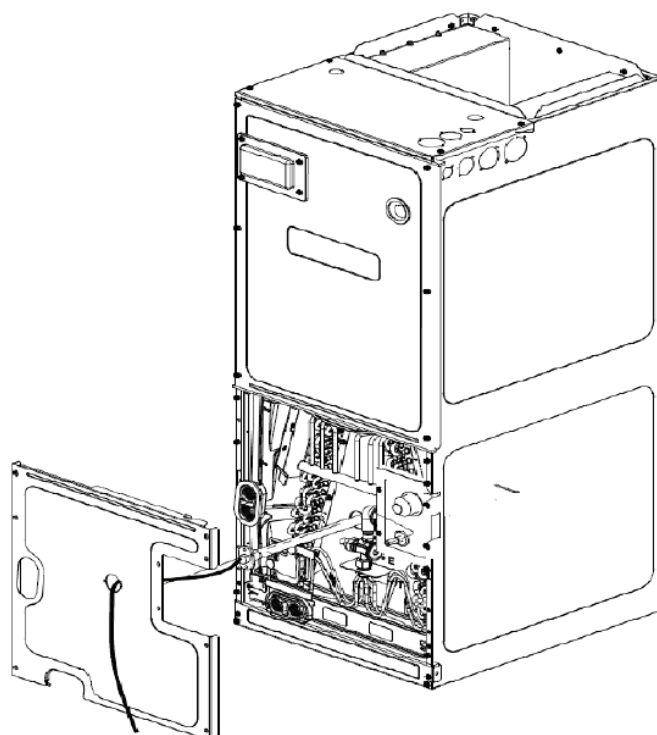


FIGURE 22

Model Size	Kit #	Parts				
AMST24B	UVPK01	DRAIN PAN MAIN	DRAIN PAN SIDE	N/A	CONDENSATE COLLECTOR, FRONT	CONDENSATE COLLECTOR, BACK
AMST30B	UVPK02					
AMST36B	UVPK02					
AMST36C	UVPK03					
AMST42C	UVPK04					
AMST48C	UVPK05					
AMST60D	UVPK07					

TABLE 19

Motors

Indoor and outdoor fan motors are permanently lubricated and do not require additional oiling.

	WARNING
HIGH VOLTAGE DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.	
	

ALUMINUM INDOOR COIL CLEANING

(QUALIFIED SERVICER ONLY)

A2L sensor must be removed from the unit before applying coil cleaners to avoid damage and contamination. Re-assemble the A2L sensor back to the unit after done with coil cleaning process.

This unit is equipped with an aluminum tube evaporator coil. The safest way to clean the evaporator coil is to simply flush the coil with water. This cleaning practice remains as the recommended cleaning method for both copper tube and aluminum tube residential evaporator coils.

It has been determined that many coil cleaners and drain pan tablets contain corrosive chemicals that can be harmful to aluminum tube and fin evaporator coils. Even a one-time application of these corrosive chemicals can cause premature aluminum evaporator coil failure. Any cleaners that contain corrosive chemicals including, but not limited to, chlorine and hydroxides, should not be used.

An alternate cleaning method is to use one of the products listed in TP-109* to clean the coils. The cleaners listed are the only agents deemed safe and approved for use to clean round tube aluminum coils. TP-109 is also available on the web site in Partner Link > Service Toolkit.

NOTE: Ensure coils are rinsed well after use of any chemical cleaners.

	CAUTION
TO AVOID THE RISK OF EQUIPMENT DAMAGE OR FIRE, INSTALL THE SAME AMPERAGE BREAKER OR FUSE AS YOU ARE REPLACING. IF THE CIRCUIT BREAKER OR FUSE SHOULD OPEN AGAIN WITHIN THIRTY DAYS, CONTACT A QUALIFIED SERVICER TO CORRECT THE PROBLEM. IF YOU REPEATEDLY RESET THE BREAKER OR REPLACE THE FUSE WITHOUT HAVING THE PROBLEM CORRECTED, YOU RUN THE RISK OF SEVERE EQUIPMENT DAMAGE.	

BEFORE YOU CALL YOUR SERVICER

- Check the thermostat to confirm that it is properly set.
- Wait 15 minutes. Some devices in the outdoor unit or in programmable thermostats will prevent compressor operation for awhile, and then reset automatically. Also, some power companies will install devices which shut off air conditioners for several minutes on hot days. If you wait several minutes, the unit may begin operation on its own.
- Check the electrical panel for tripped circuit breakers or open fuses. Reset the circuit breakers or replace fuses as necessary.
- Check the disconnect switch near the indoor furnace or blower to confirm that it is closed.
- Check for obstructions on the outdoor unit. Confirm that it has not been covered on the sides or the top. Remove any obstruction that can be safely removed. If the unit is covered with dirt or debris, call a qualified servicer to clean it.
- Check for blockage of the indoor air inlets and outlets. Confirm that they are open and have not been blocked by objects (rugs, curtains or furniture).
- Check the filter. If it is dirty, clean or replace it.
- Listen for any unusual noise(s), other than normal operating noise, that might be coming from the outdoor unit. If you hear unusual noise(s) coming from the unit, call a qualified servicer.

HOMEOWNER SUPPORT
DAIKIN COMFORT TECHNOLOGIES
MANUFACTURING, INC.
19001 KERMIER ROAD
WALLER, TEXAS 77484
(855) 770-5678

20 Start-Up Checklist

Air Handler / Coil			
		Model Number	_____
		Serial Number	_____
ELECTRICAL			
Line Voltage (Measure L1 and L2 Voltage)	L1 - L2	_____	
Secondary Voltage (Measure Transformer Output Voltage)	R - C	_____	
Blower Amps		_____	
Heat Strip 1 - Amps		_____	
Heat Strip 2 - Amps		_____	
BLOWER EXTERNAL STATIC PRESSURE			
Return Air Static Pressure		_____	IN. W.C.
Supply Air Static Pressure		_____	IN. W.C.
Total External Static Pressure (Ignoring +/- from the reading above, add total here)		_____	IN. W.C.
TEMPERATURES			
Return Air Temperature (Dry bulb / Wet bulb)		_____ DB °F	_____ WB °F
Cooling Supply Air Temperature (Dry bulb / Wet bulb)		_____ DB °F	_____ WB °F
Heating Supply Air Temperature		_____ DB °F	
Temperature Rise		_____ DB °F	
Delta T (Difference between Supply and Return Temperatures)		_____ DB °F	
Air Handler / Coil - (Inverter Matched)			
INVERTER AH / COIL ONLY			
Check EEV and EEV wiring is secure (no adjustment required)		_____	
Additional Checks			
Check wire routings for any rubbing		_____	
Check product for proper draining		_____	
Check screw tightness on blower wheel		_____	
Check factory wiring and wire connections		_____	
Check product for proper clearances as noted by installation instructions		_____	
°F to °C formula: (°F - 32) divided by 1.8 = °C °C to °F formula: (°C multiplied by 1.8) + 32 = °F			

CUSTOMER FEEDBACK

We are very interested in all product comments.

Please fill out the feedback form on one of the following links:

Daikin Products: (<https://daikincomfort.com/contact-us>)

Goodman® Brand Products: (<http://www.goodmanmfg.com/about/contact-us>).

Amana® Brand Products: (<http://www.amana-hac.com/about-us/contact-us>).

You can also scan the QR code on the right for the product brand you purchased to be directed to the feedback page.



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19001 Kermier Rd., Waller, TX 77484

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